

Intro to ML

Naïve Bayes

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ITMO, fall 2026

Outline

1. ML and AI overview
2. What is Machine Learning
3. Thesaurus and notation
4. Datasets
5. Exploratory data analysis
6. Maximum Likelihood Estimation
7. Some Machine Learning problems
8. Naïve Bayes classifier
9. Machine learning libraries

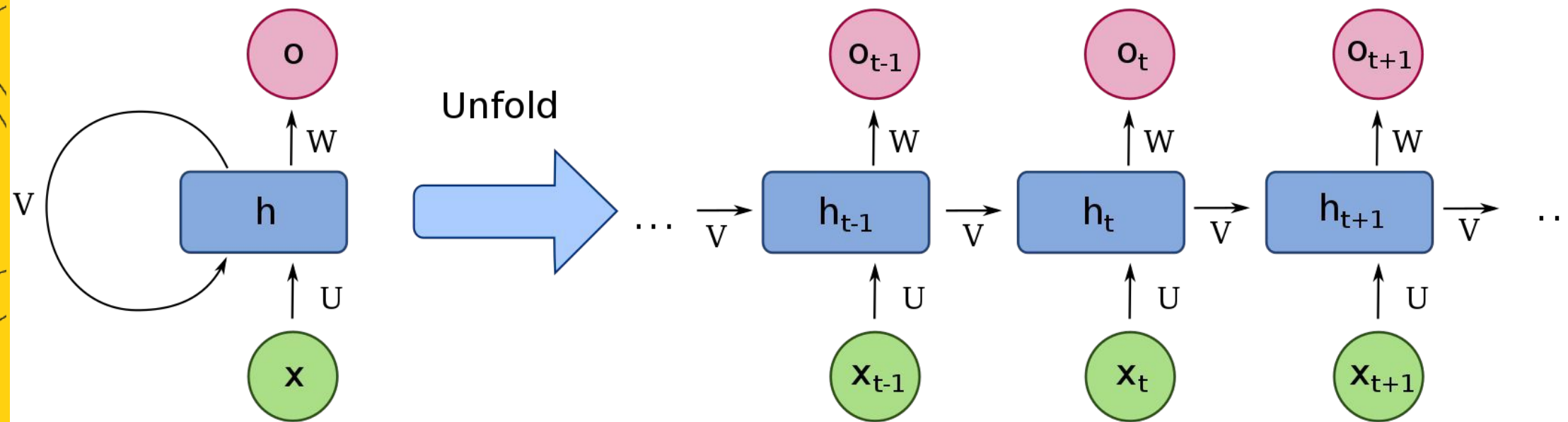
ML and AI overview

girafe
ai

01



Natural Language Processing



Basics:

- Text vectorization
- Language models
- Recurrent Neural Networks
- Attention module



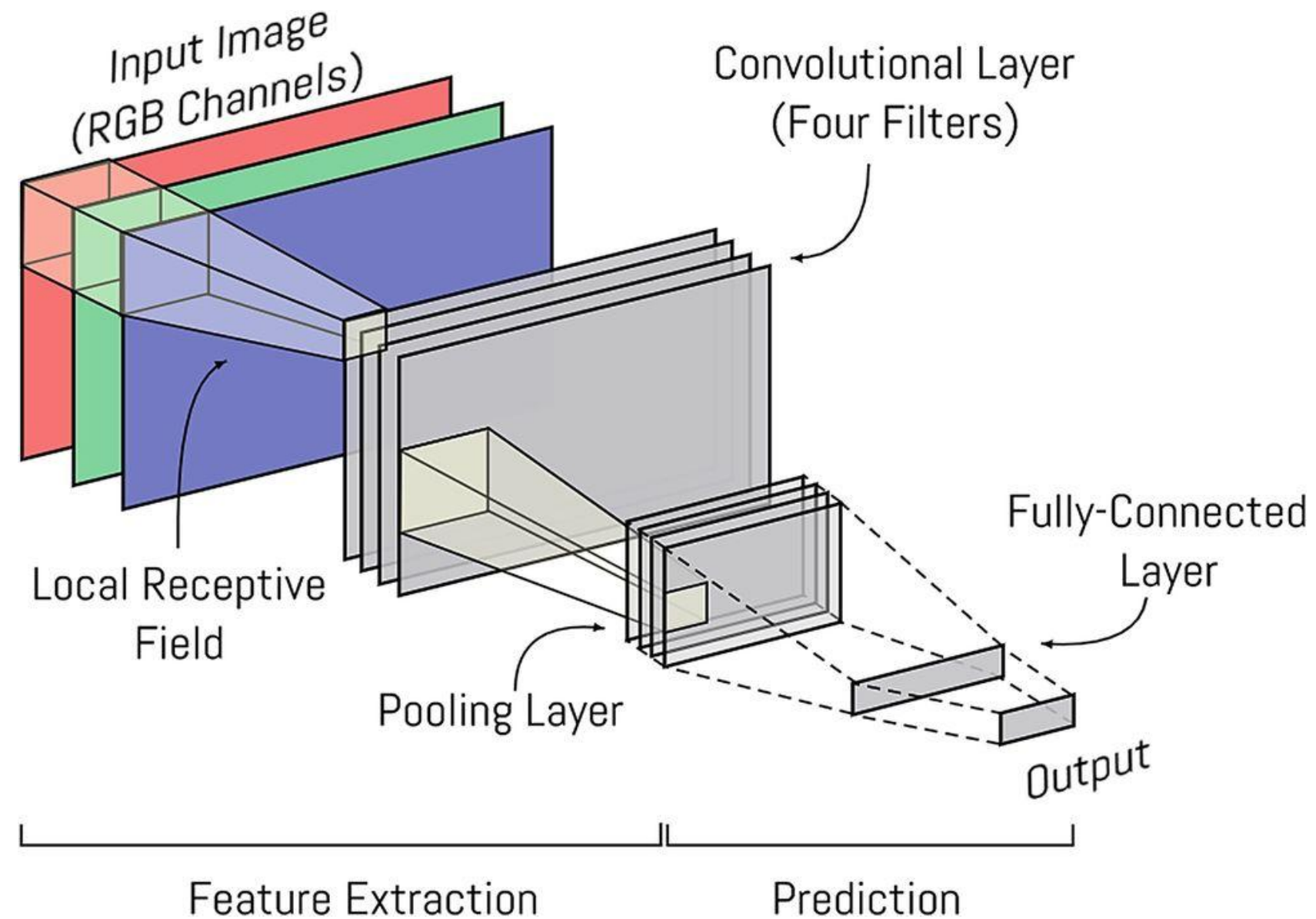
Natural Language Processing

Some achievements:

- Machine translation
- Texts classification
- Texts generation



Computer Vision



Basics:

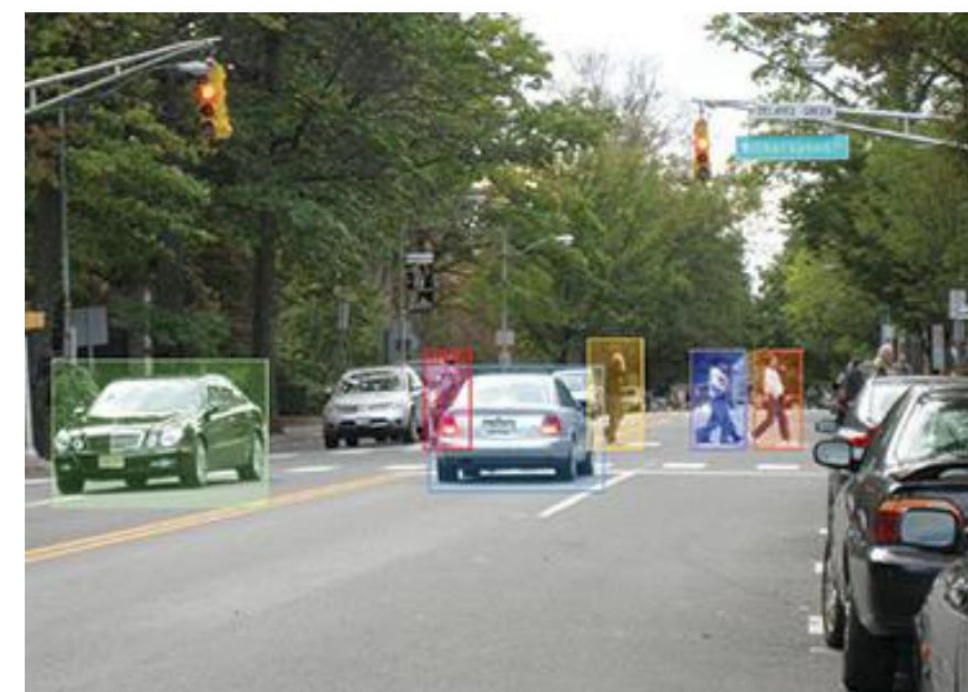
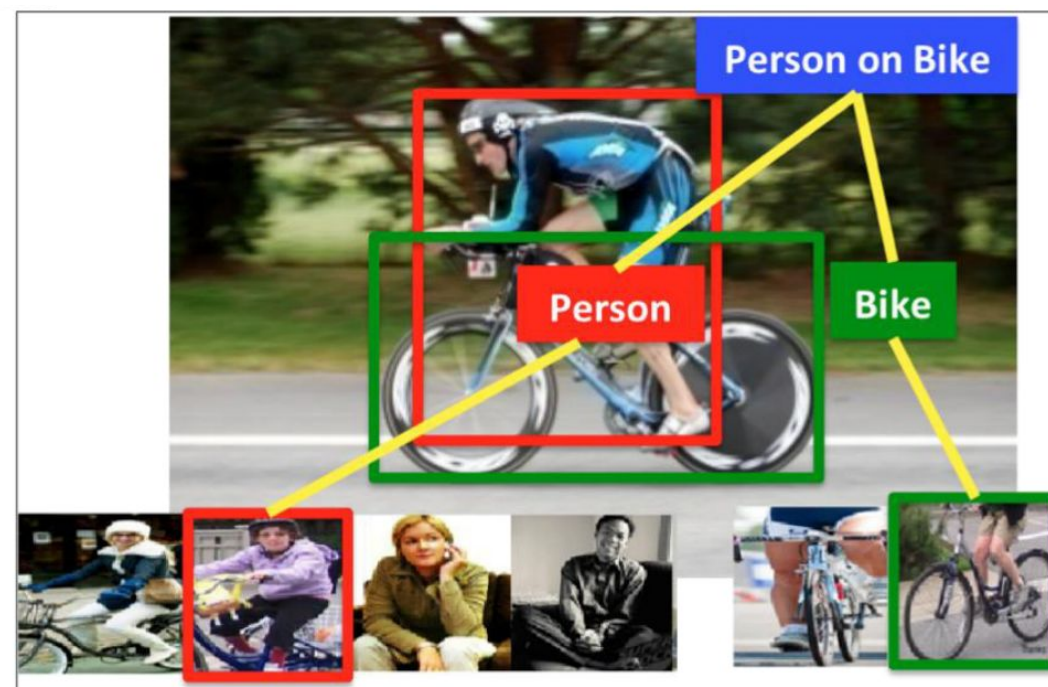
- Classical CV (filters, border detectors)
- Convolutional Neural Networks

Computer Vision



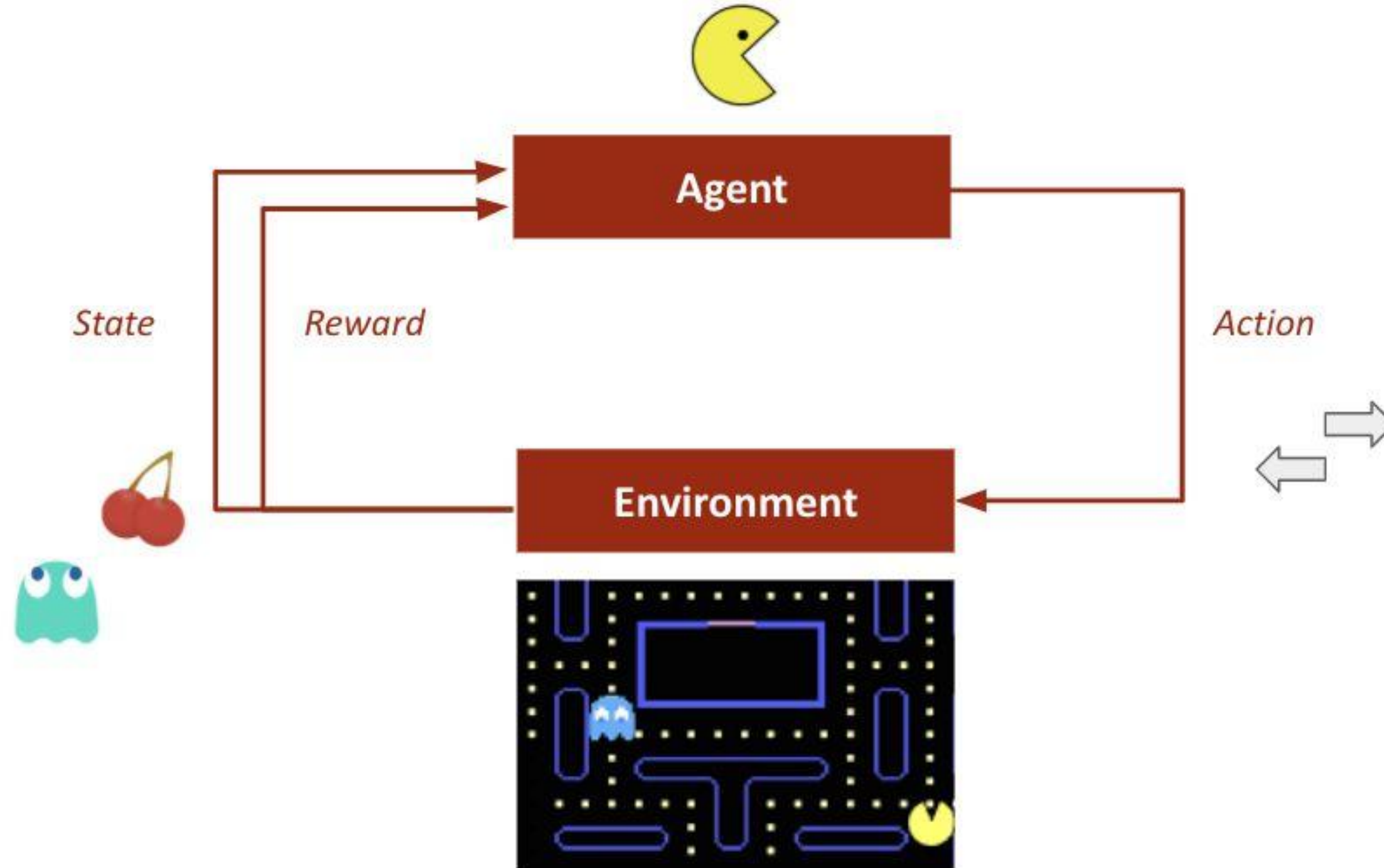
Some achievements:

- Object detection
- Semantic segmentation
- Generative models





Reinforcement Learning



Basics:

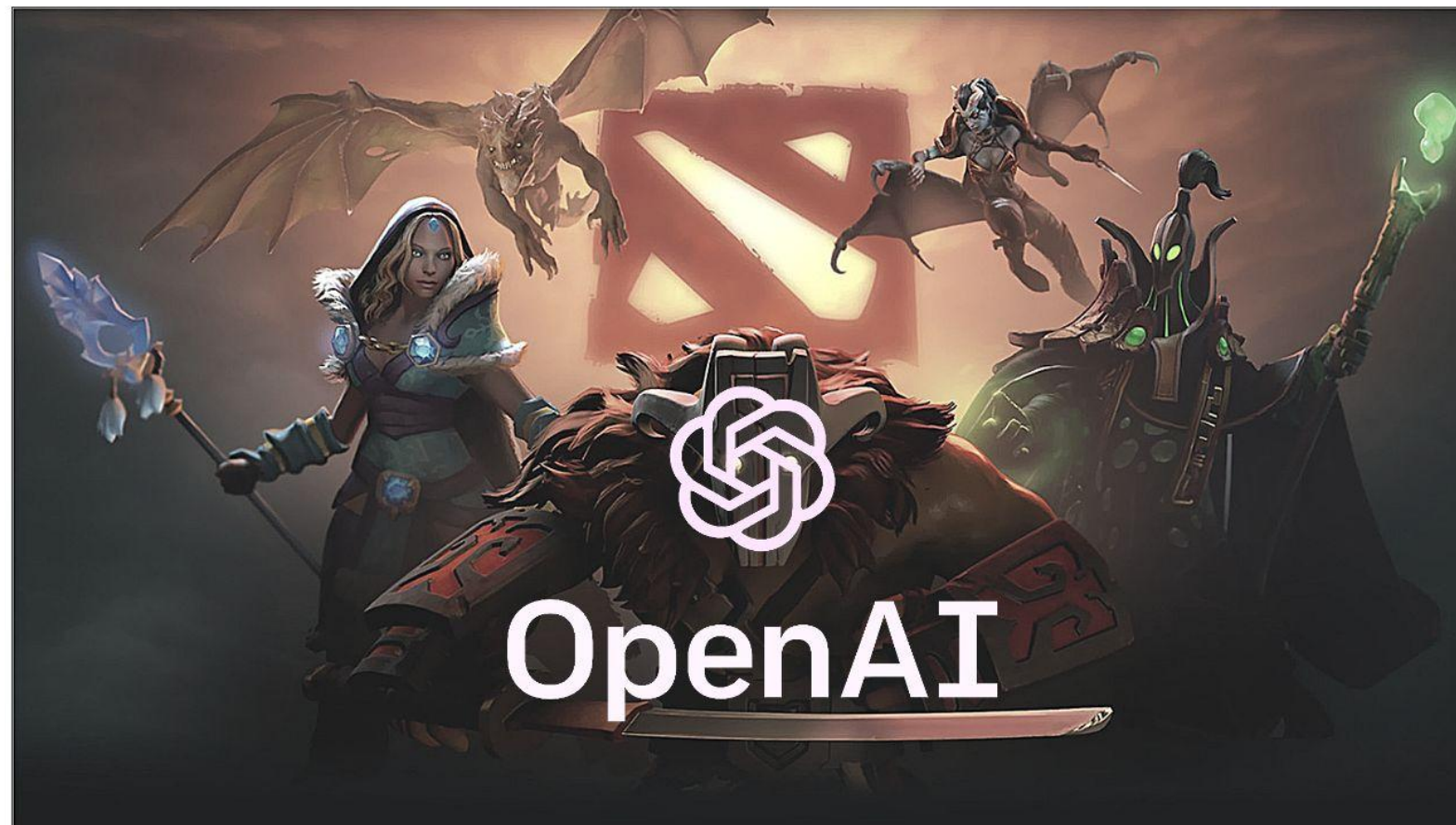
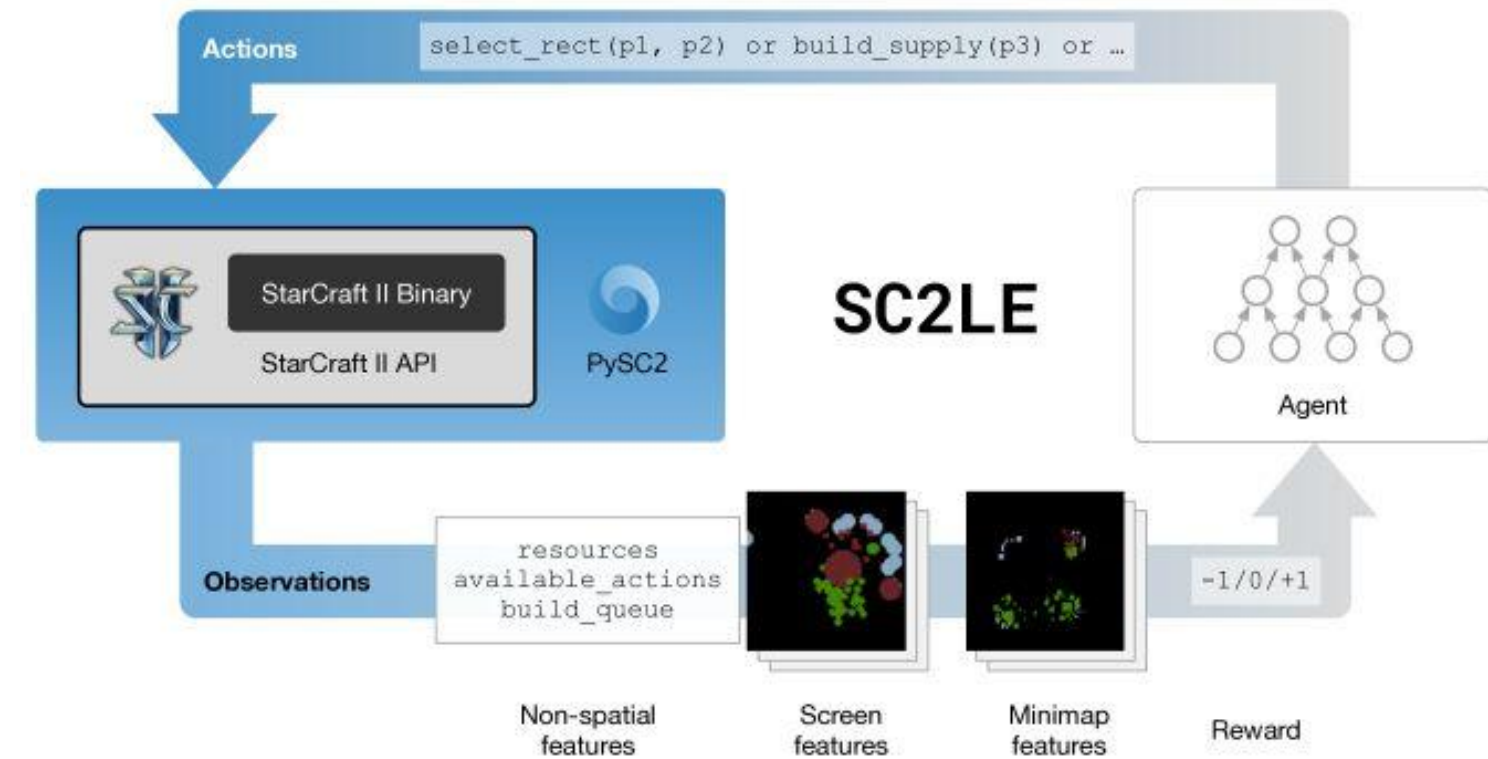
- Q-learning
- DQN
- REINFORCE



Reinforcement Learning

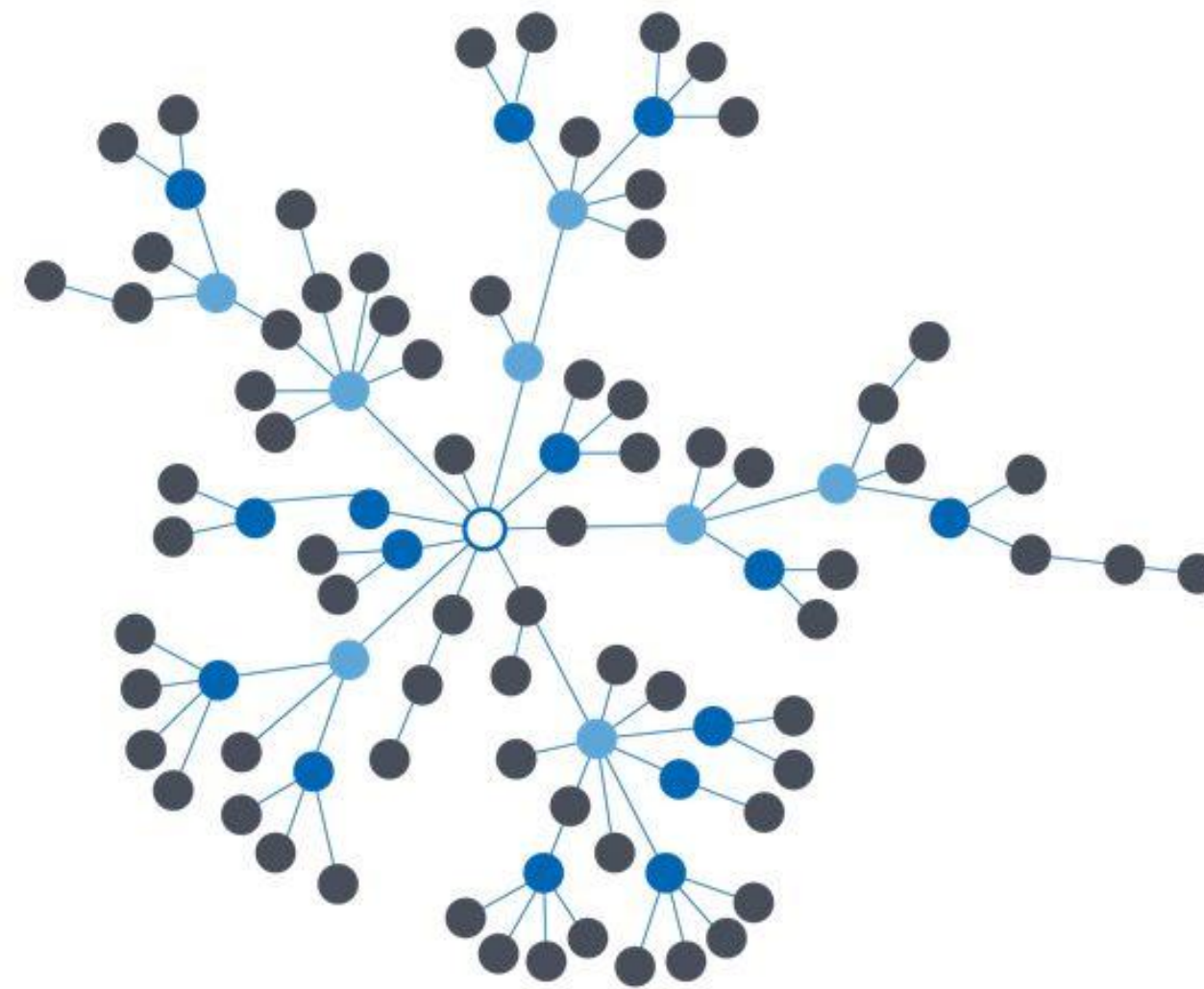
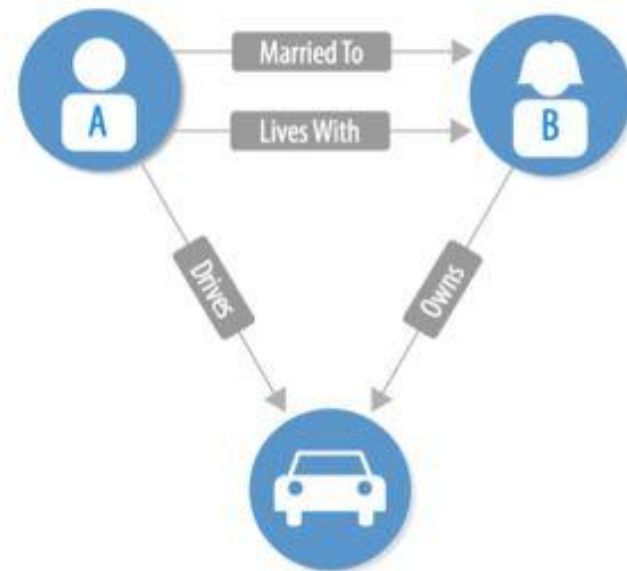
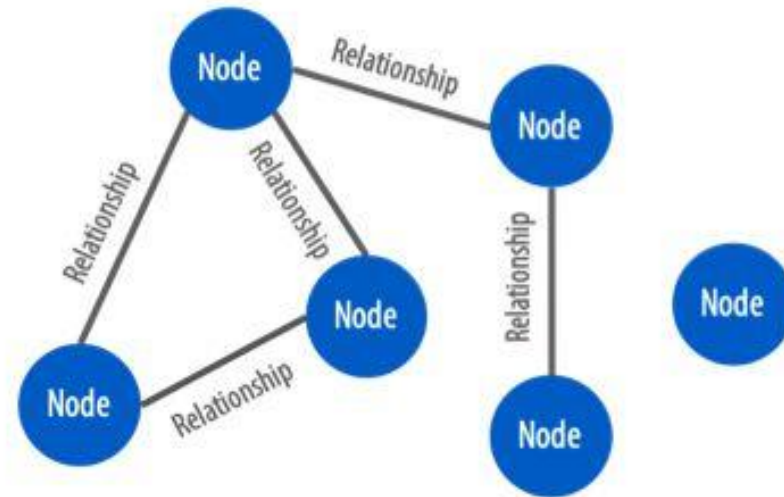
Achievements:

- Alpha Go
- OpenAI Five
- DeepMind Star Craft 2





Machine Learning on Graphs



Basics:

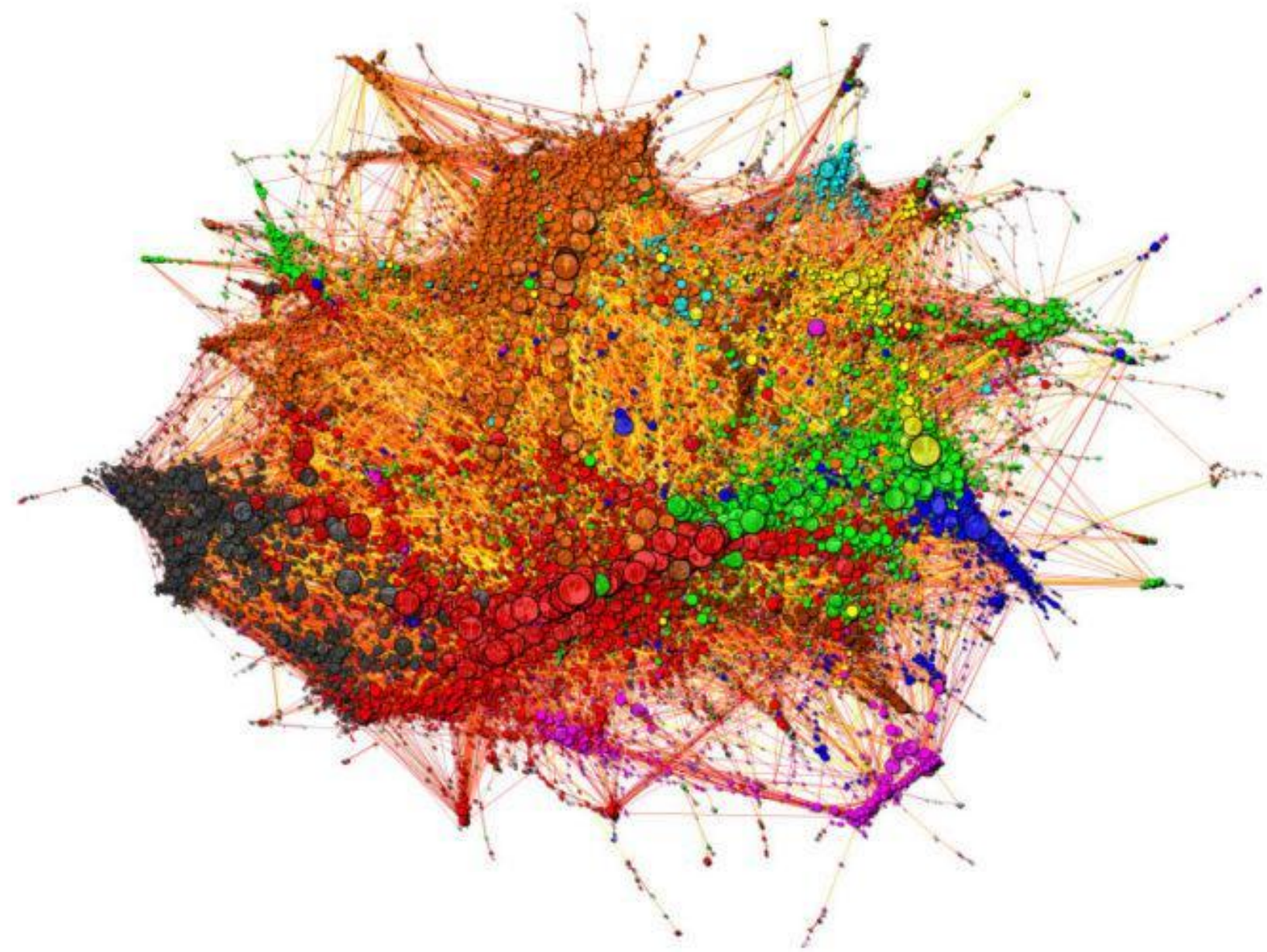
- Random graphs
- Small world model
- Graphs convolutions

Machine Learning on Graphs

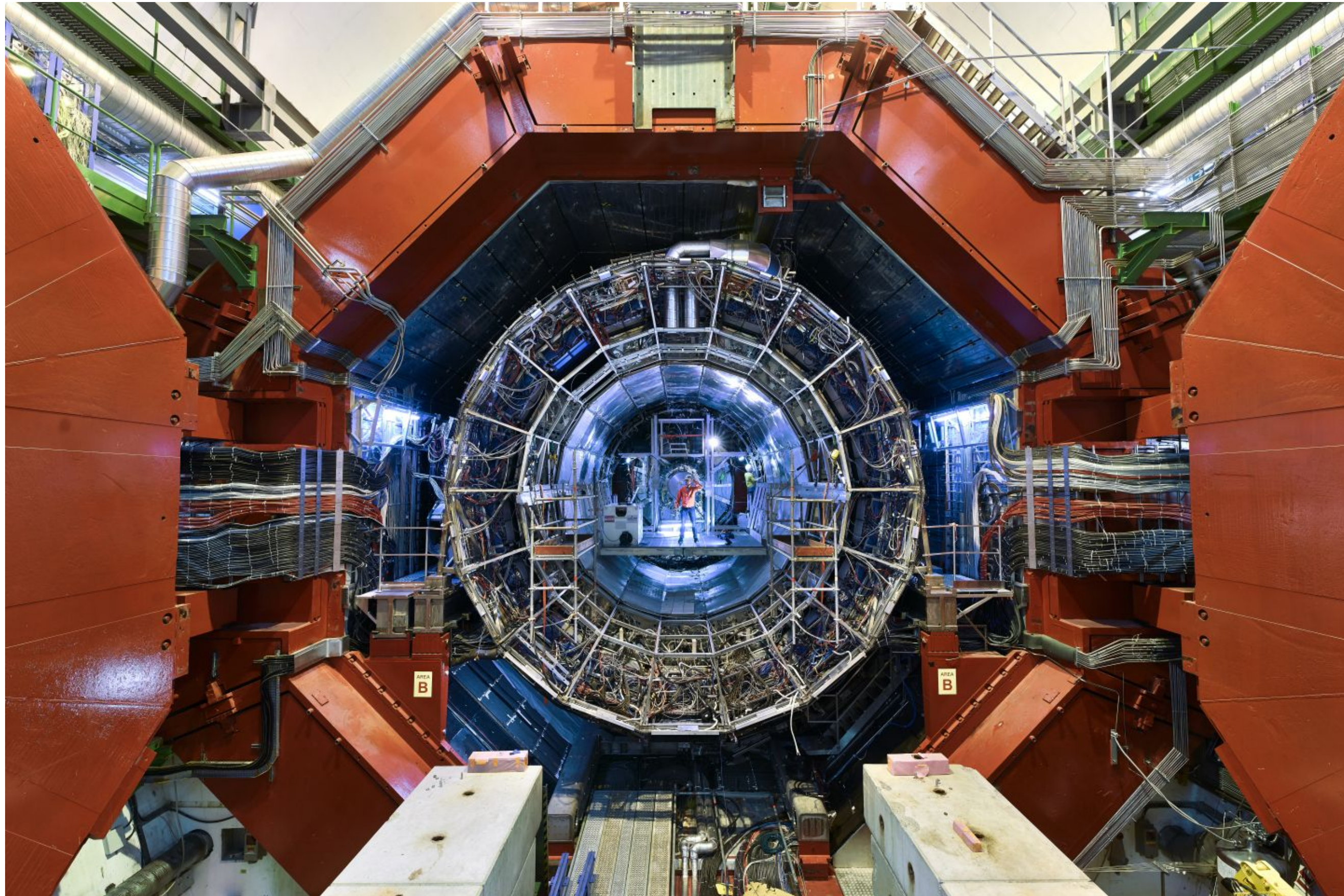


Some achievements:

- Communities detection
- Recommender systems



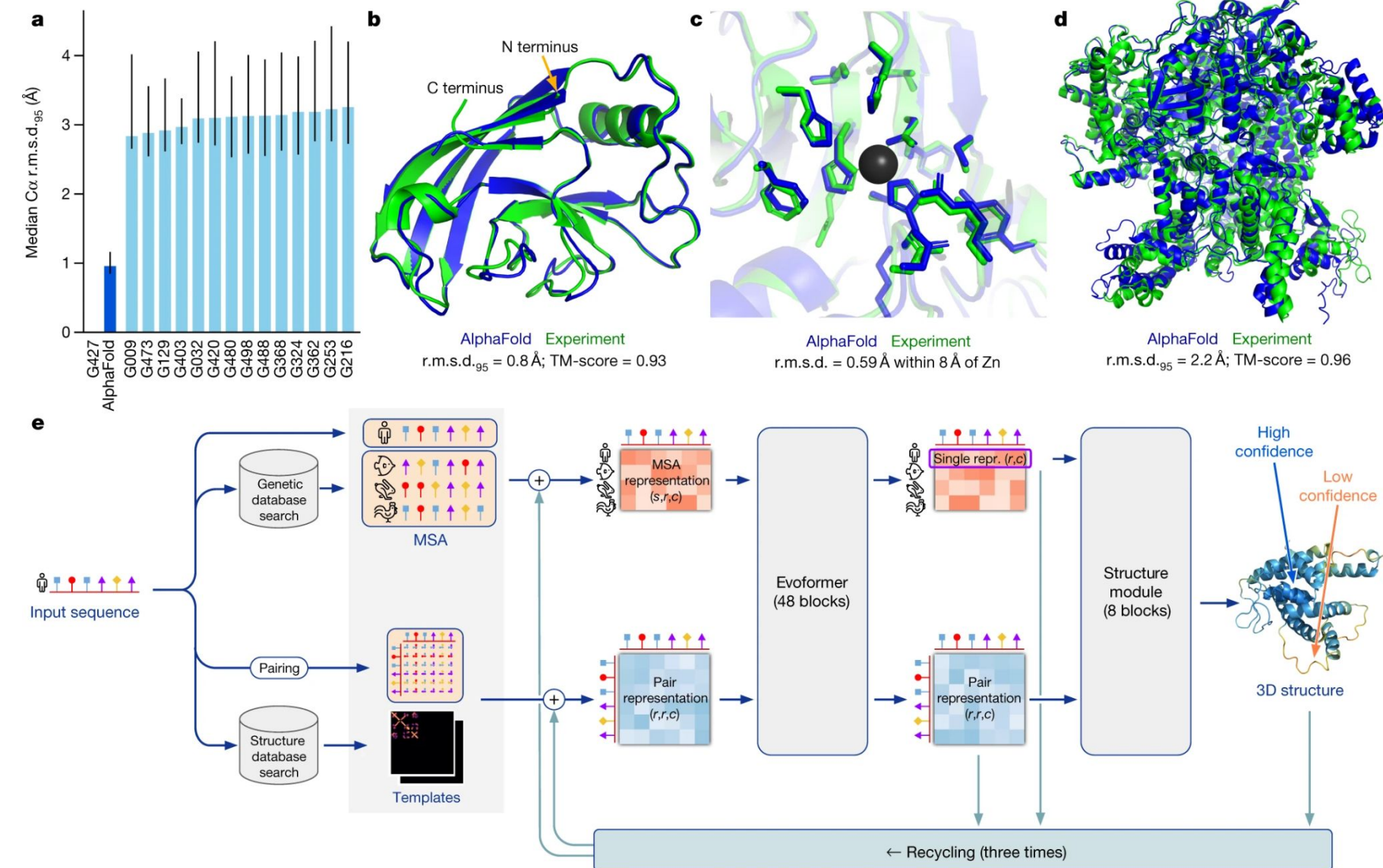
Machine Learning for Science



Nobel Prizes won for ML



- Chemistry, 2024
 - [Highly accurate protein structure prediction with AlphaFold, Jumper et al., Nature, 2021](#)
- Physics, 2024
 - [Neural networks and physical systems with emergent collective computational abilities, J J Hopfield, PNAS, 1982](#)
 - [Learning representations by back-propagating errors, David E. Rumelhart, Geoffrey E. Hinton, Ronald J. Williams, Nature, 1986](#)



Physics Informed Machine Learning



Great overview of physics informed models

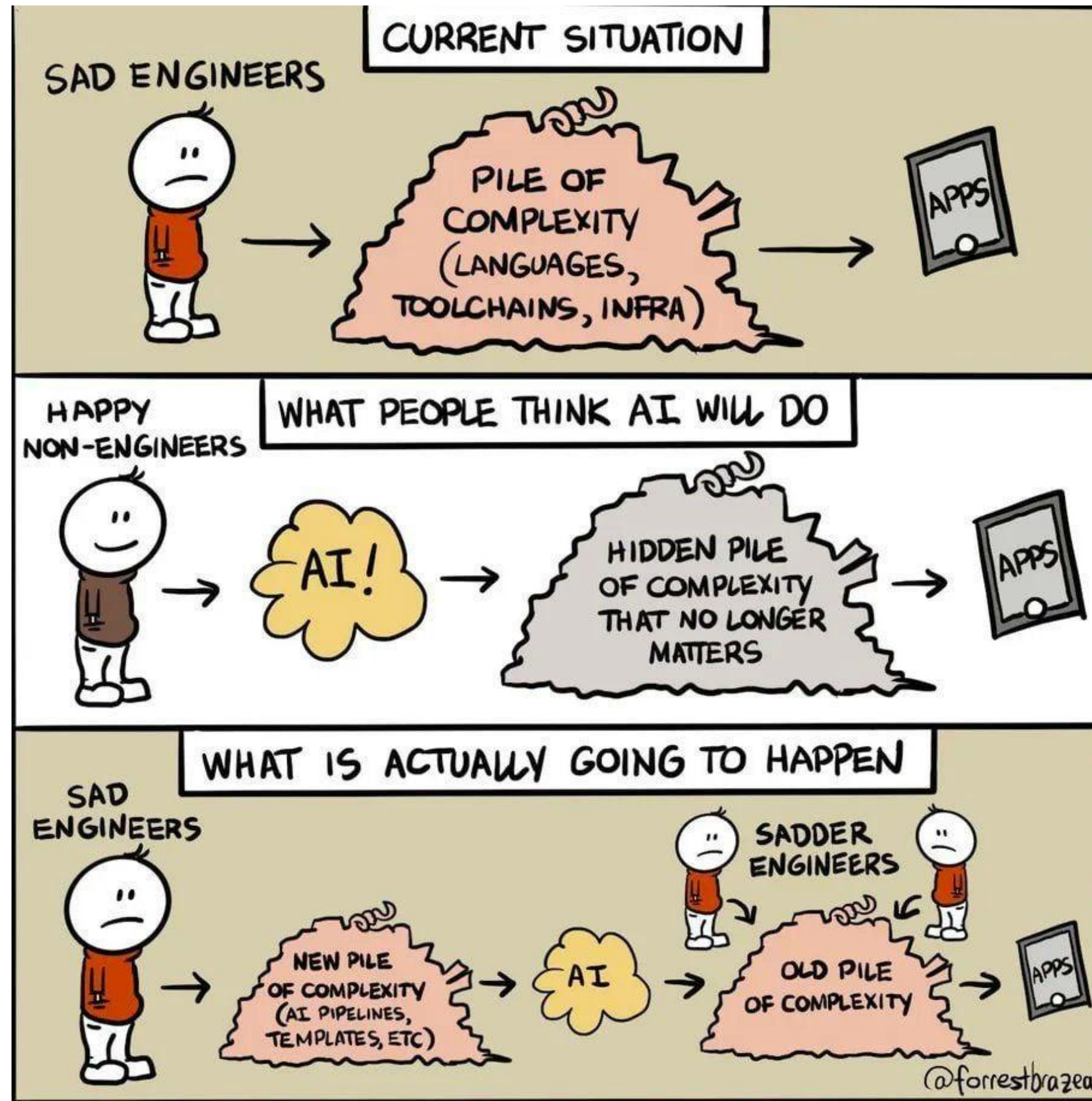
<https://www.youtube.com/watch?v=JoFW2uSd3Uo&list=PLMrJAKhleNNQ0BaKuBKY43k4xMo6NSbBa>

PHYSICS INFORMED MACHINE LEARNING

$\ddot{\theta} = -\frac{g}{L} \sin(\theta) - \delta \dot{\theta}$

PHYSICAL MODELS FROM DATA

What is AI and ML for you



What is Machine Learning?

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02

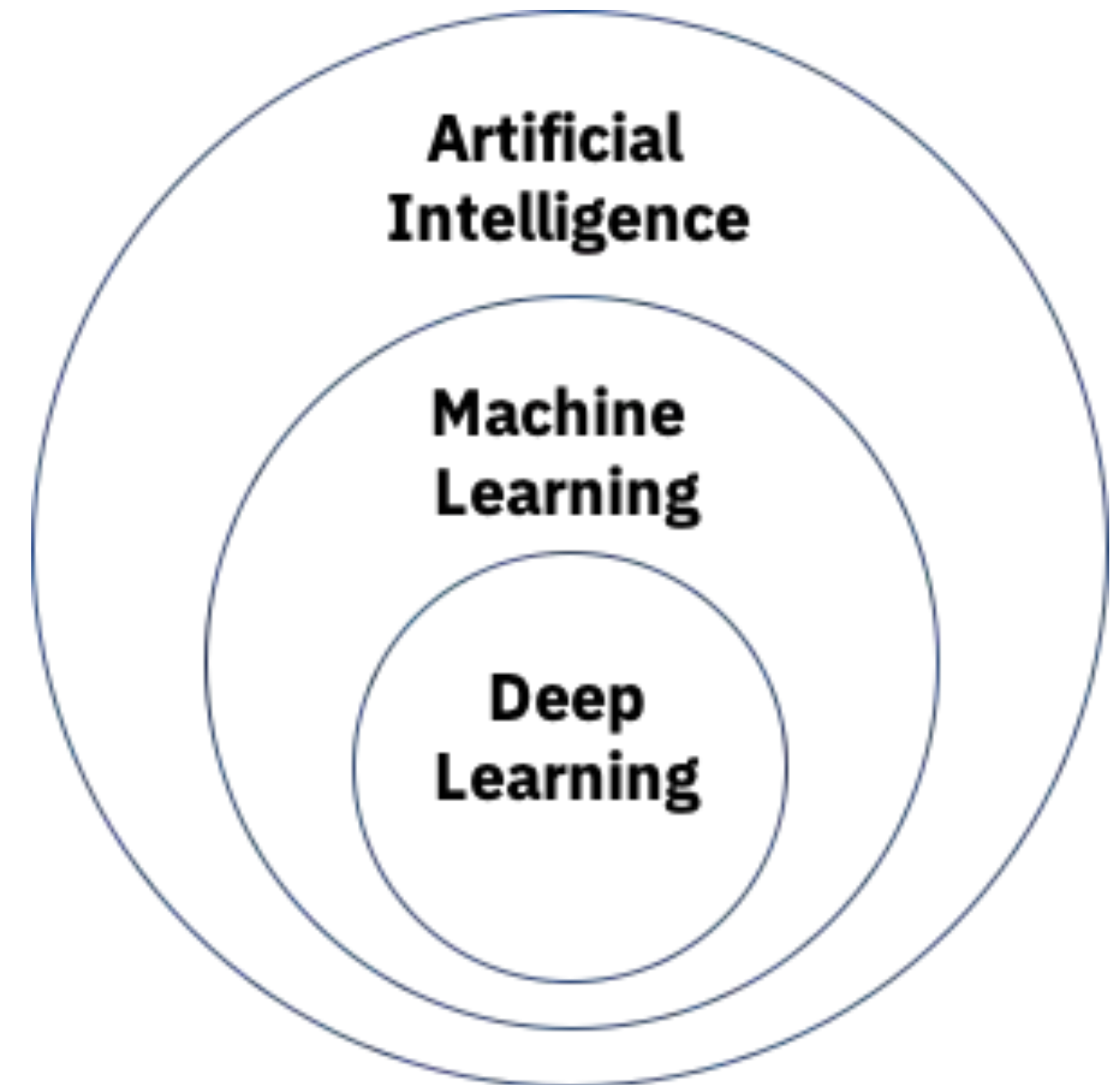


Terms around ML



There are different words that occur in similar contexts, but mean different things:

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Data Science
- Big Data



Artificial Intelligence



Any system that perceives its environment and takes actions that maximize its chances of achieving its goals. [Wikipedia](#)



The electric kettle is already part of Skynet!

Machine Learning



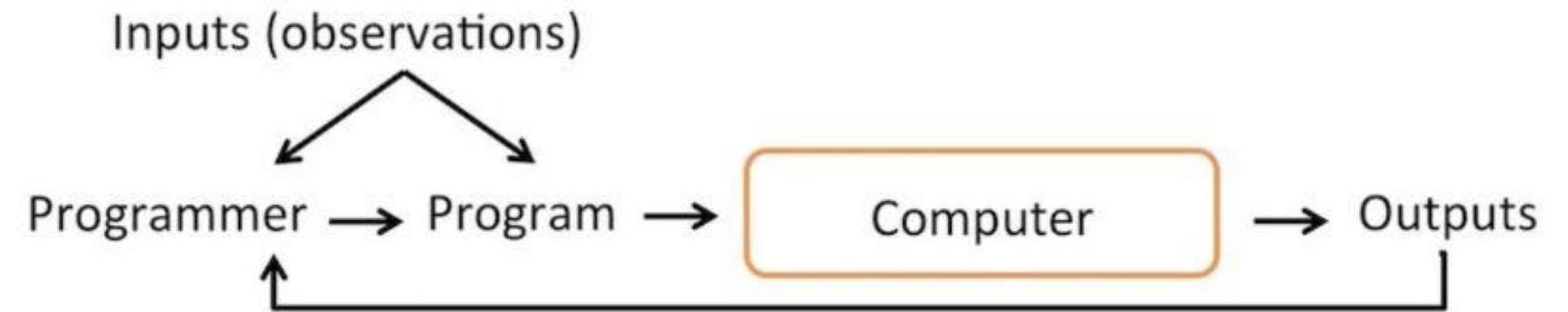
The study of computer algorithms that improve automatically through experience and by the use of data.

[Wikipedia](#)

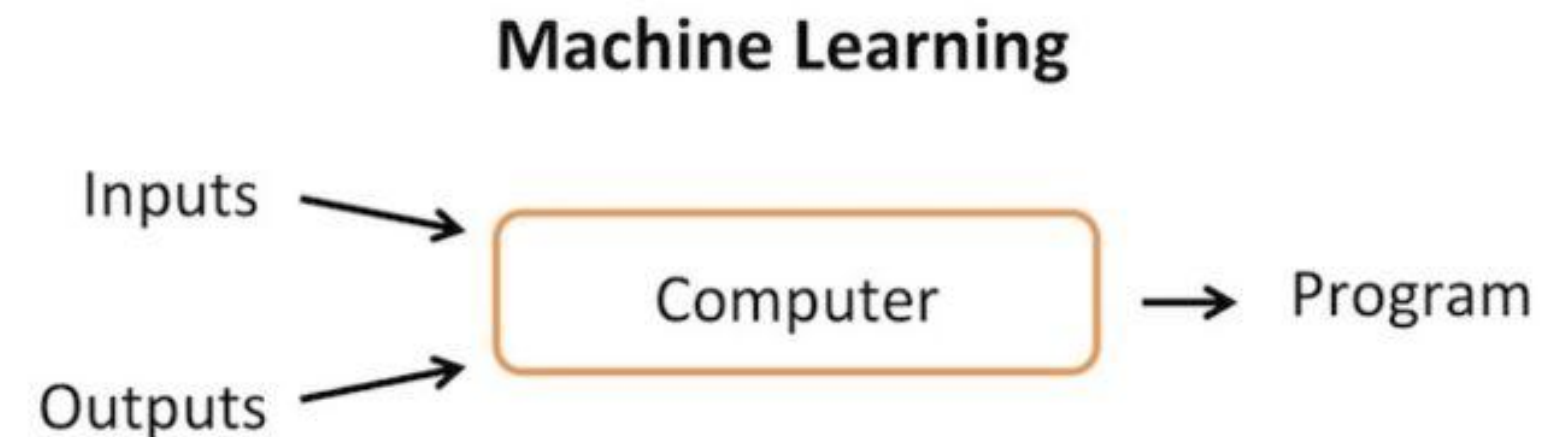
The most important distinguishing feature of machine learning is the statistical nature of the models.

Andrej Karpathy on software development

<https://www.youtube.com/watch?v=LCemiRjPEtQ>



Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed
– Arthur Samuel (1959)



Software 2.0



Talk by Andrej Karpathy <https://www.youtube.com/watch?v=y57wwucbXR8>



Andrej Karpathy

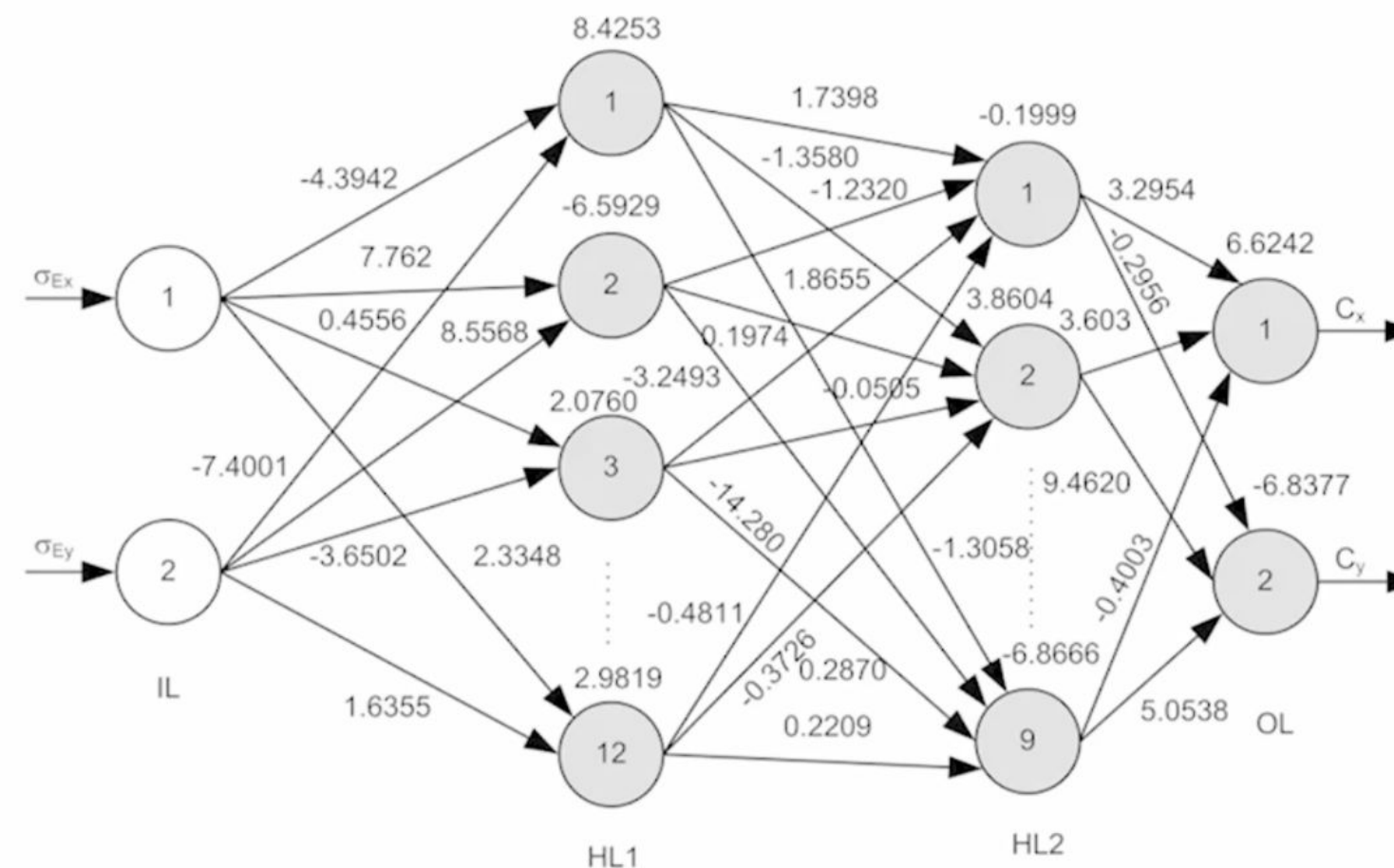
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Software 1.0 = code



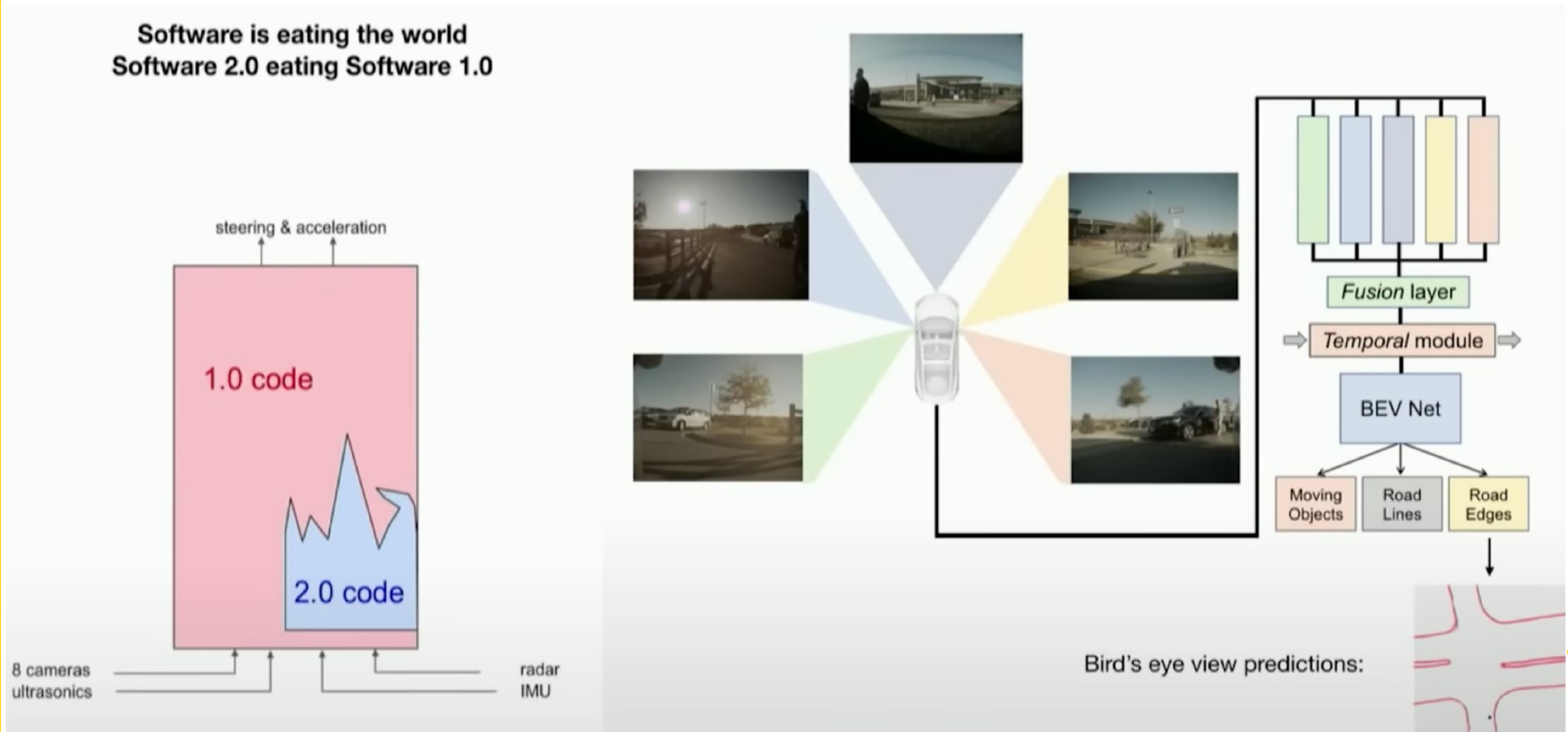
Software 2.0 = weights



Software 2.0 growth



<https://karpathy.medium.com/software-2-0-a64152b37c35>



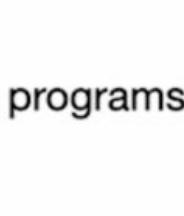
Software 3.0



Talk on Software 3.0 <https://www.youtube.com/watch?v=LCEmiRjPEtQ>

Software 1.0

computer code



computer



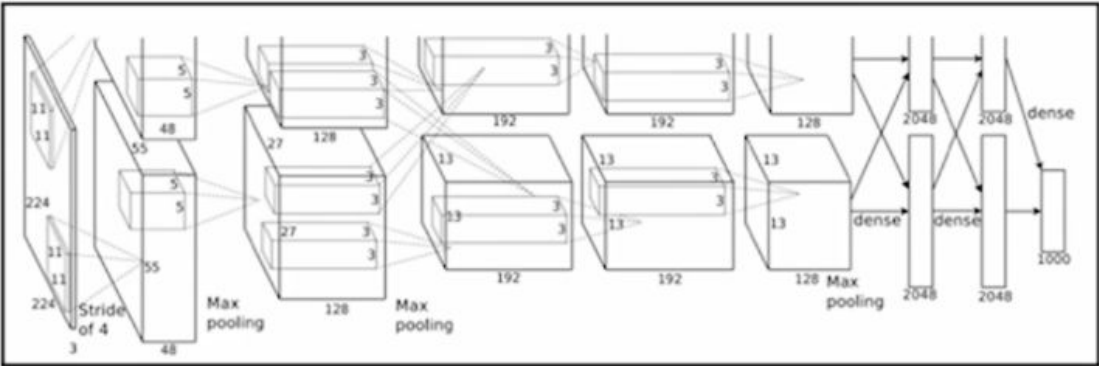
became programmable in ~1940s

Software 2.0

weights



neural net



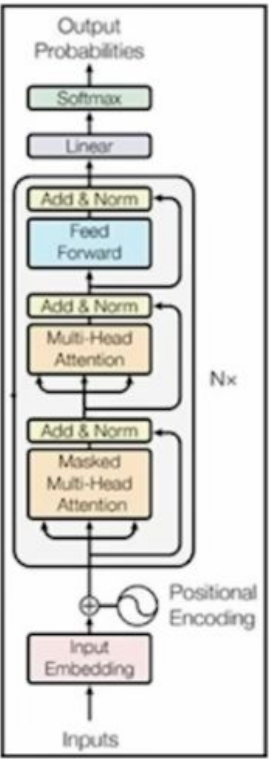
fixed function neural net
e.g. AlexNet: for image recognition (~2012)

Software 3.0

prompts



LLM



~2019

LLM = programmable neural net!

Example: Sentiment classification



Software 1.0

```
python
def simple_sentiment(review: str) -> str:
    """Return 'positive' or 'negative' based on a tiny keyword lexicon."""
    positive = {
        "good", "great", "excellent", "amazing", "wonderful", "fantastic",
        "awesome", "loved", "love", "like", "enjoyed", "superb", "delightful"
    }
    negative = {
        "bad", "terrible", "awful", "poor", "boring", "hate", "hated",
        "dislike", "worst", "dull", "disappointing", "mediocre"
    }

    score = 0
    for word in review.lower().split():
        w = word.strip(".,!?:") # crude token clean-up
        if w in positive:
            score += 1
        elif w in negative:
            score -= 1

    return "positive" if score >= 0 else "negative"
Copy
```

Software 2.0

10,000 positive examples
10,000 negative examples
encoding (e.g. bag of words)

train binary classifier

parameters

Software 3.0

You are a sentiment classifier. For every review that appears between the tags

<REVIEW> ... </REVIEW>, respond with **exactly one word**, either POSITIVE or NEGATIVE (all-caps, no punctuation, no extra text).

Example 1

<REVIEW>I absolutely loved this film—the characters were engaging and the ending was perfect.</REVIEW>

POSITIVE

Example 2

<REVIEW>The plot was incoherent and the acting felt forced; I regret watching it.</REVIEW>

NEGATIVE

Example 3

<REVIEW>An energetic soundtrack and solid visuals almost save it, but the story drags and the jokes fall flat.</REVIEW>

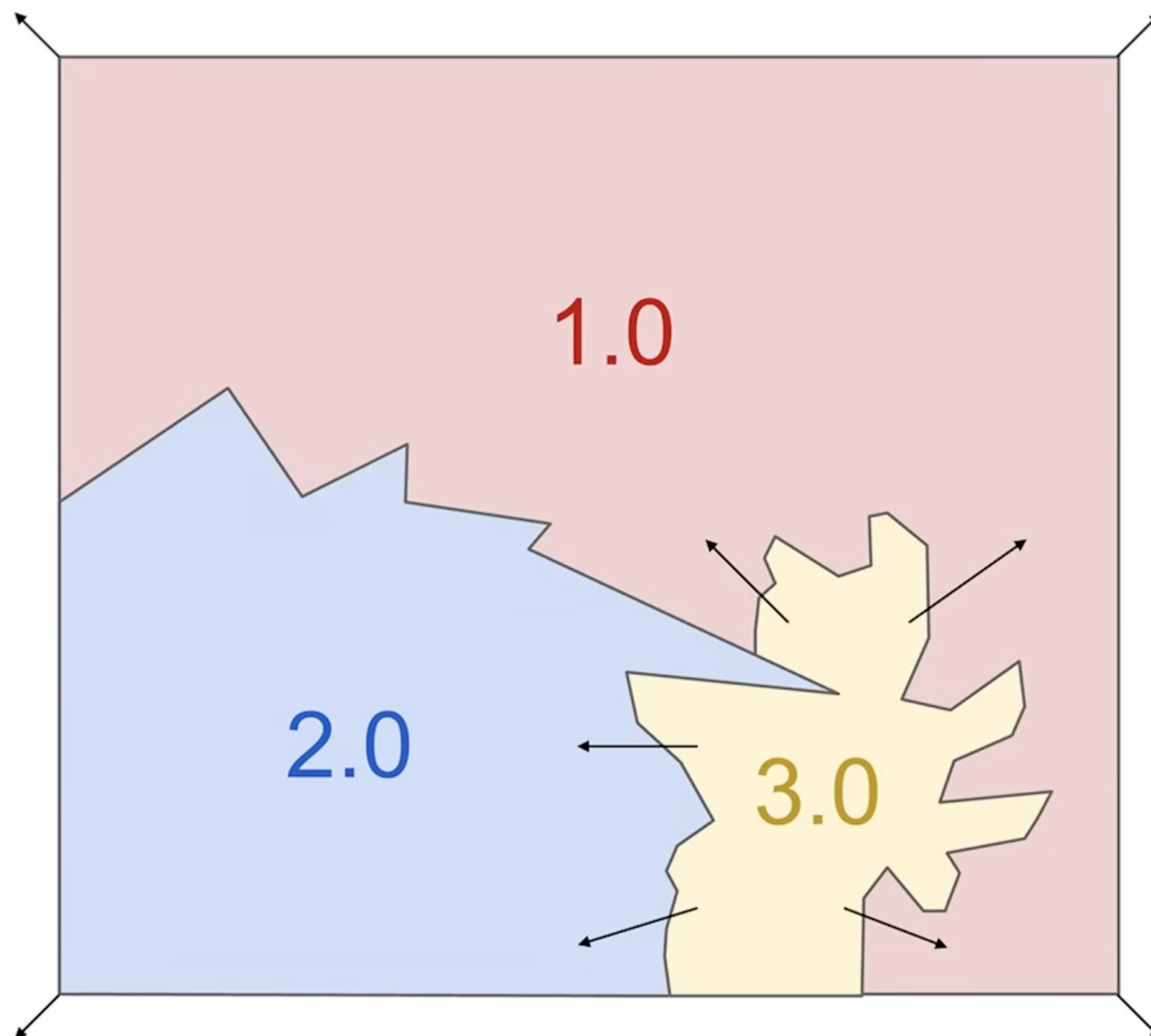
NEGATIVE

Now classify the next review.

Software conquest



A huge amount of Software will be (re-)written.



Across many applications areas,
we'll be left with a choice of using:

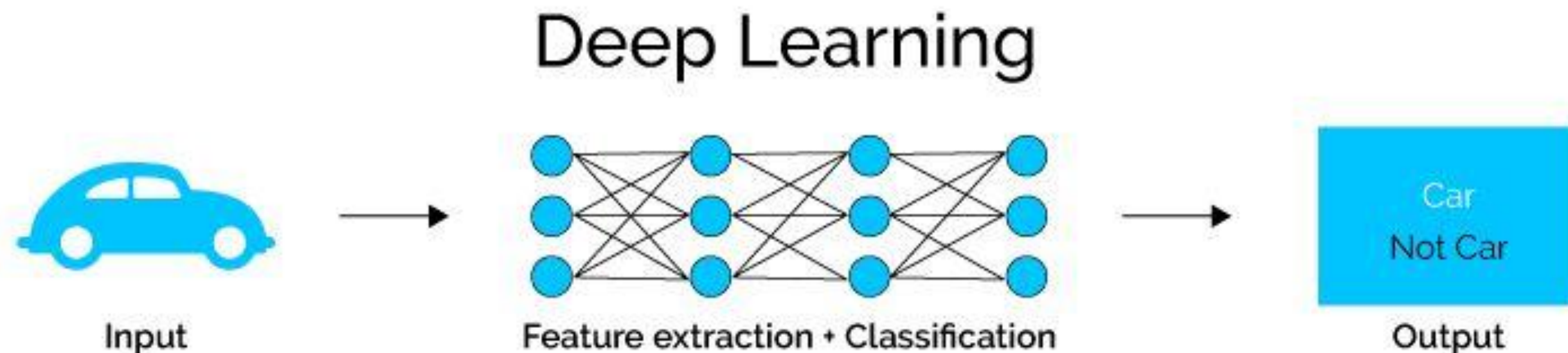
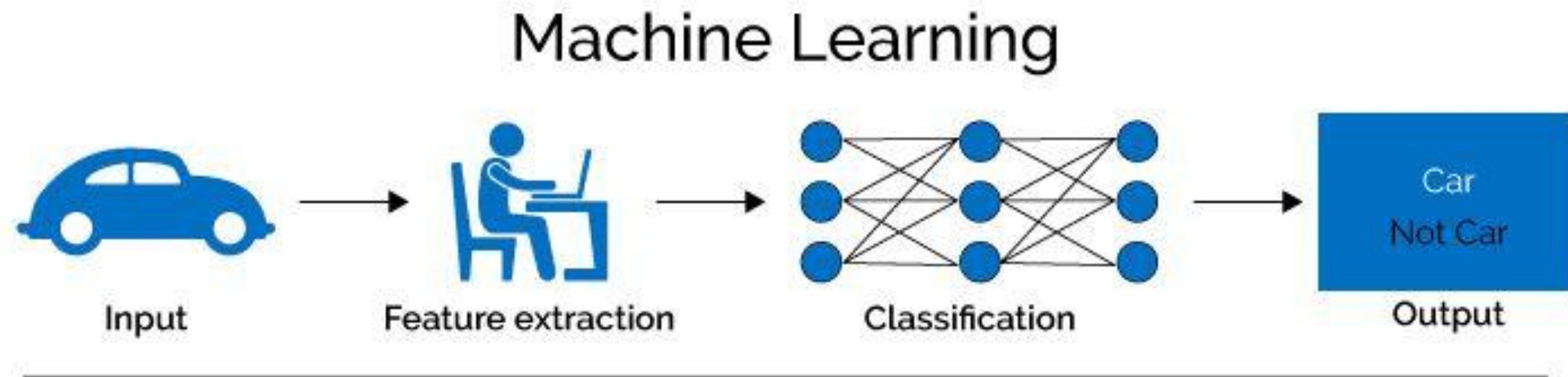
- 90% accurate model we understand
- 99% accurate model we don't

Deep Learning



A specific type of models that has shown high results in many applied tasks in the last decade

One of the key success factors is compatibility with modern computing systems (GPUs), simplicity and flexibility of the architecture of a particular model



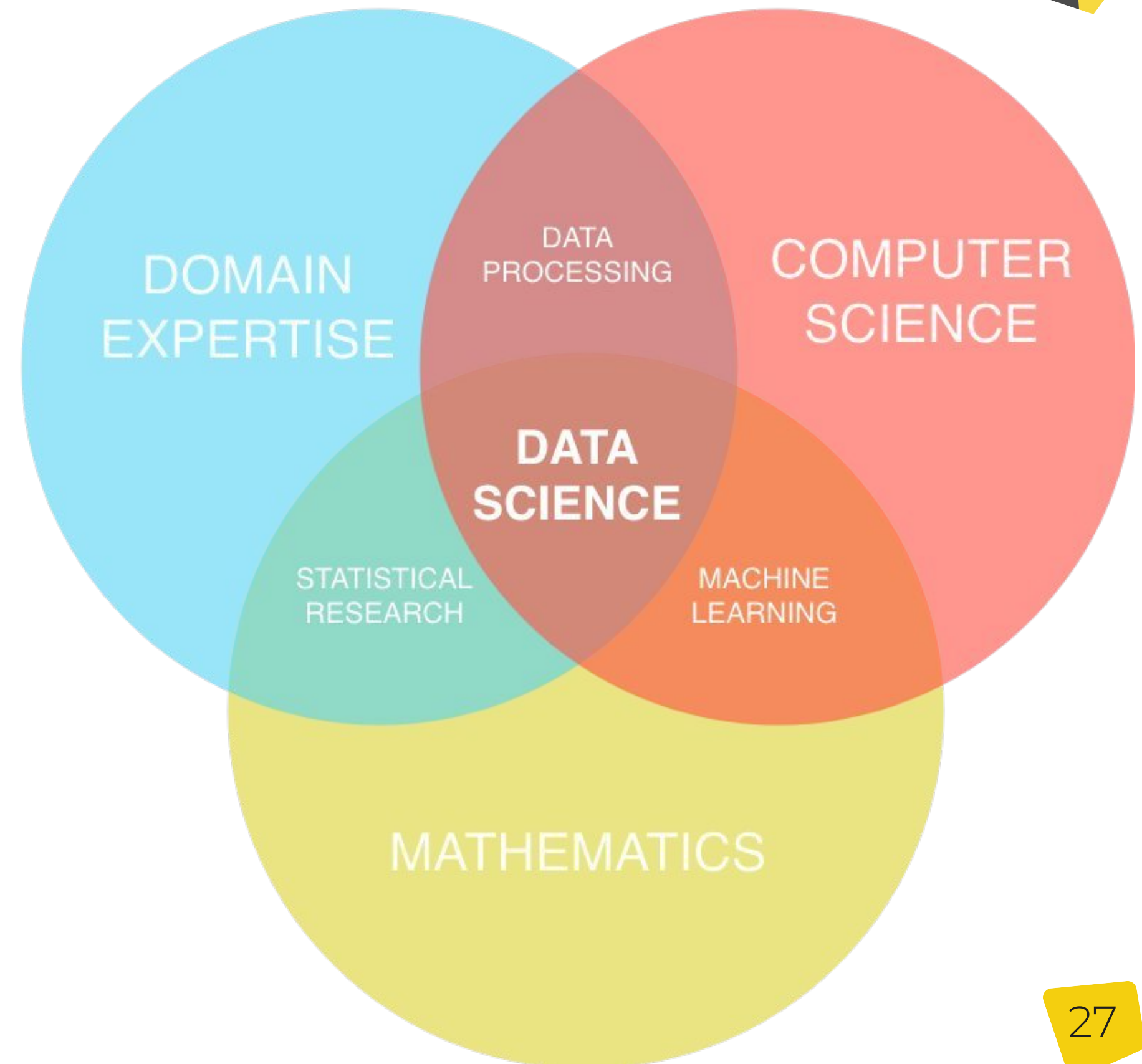
Data Science



An interdisciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data, and apply knowledge and actionable insights from data across a broad range of application domains.

[Wikipedia](#)

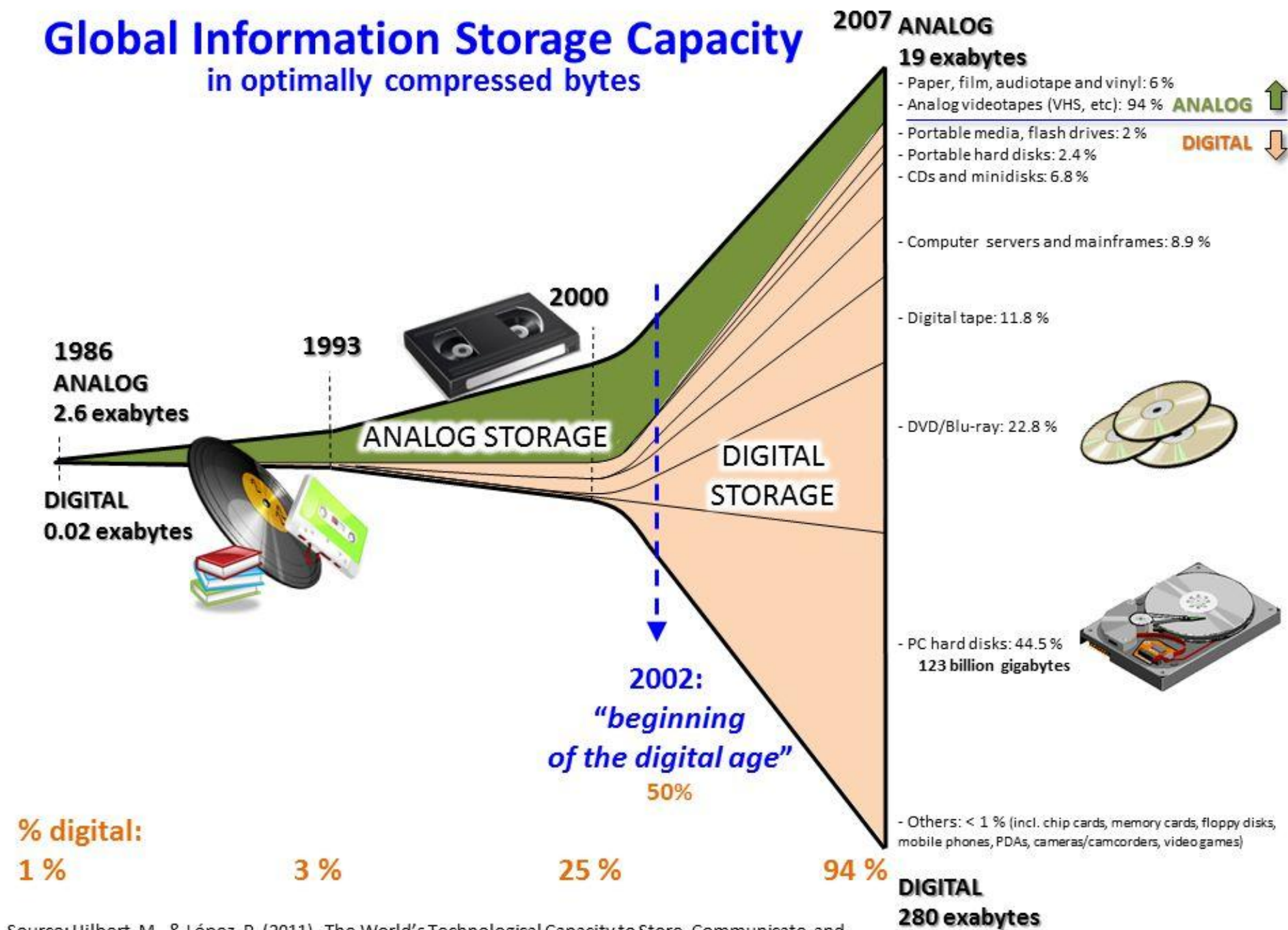
That is, it is the application of mathematical methods to data to extract useful information (so-called insights)



Big Data



Global Information Storage Capacity in optimally compressed bytes

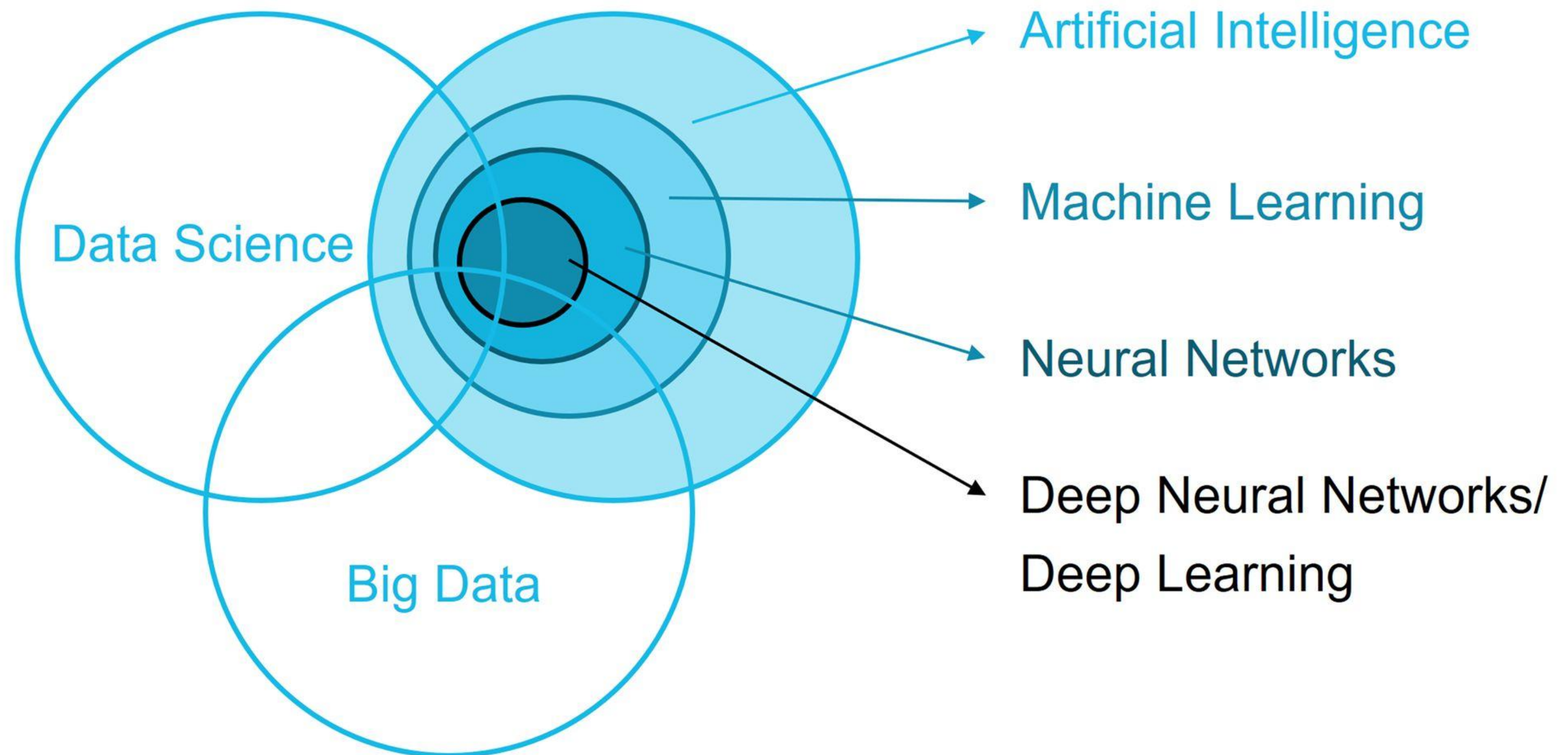


Source: Hilbert, M., & López, P. (2011). The World's Technological Capacity to Store, Communicate, and Compute Information. *Science*, 332(6025), 60–65. <http://www.martinhilbert.net/WorldInfoCapacity.html>

A field that treats ways to analyze, systematically extract information from, or otherwise deal with data sets that are too large or complex to be dealt with by traditional data-processing application software. [Wikipedia](https://en.wikipedia.org/wiki/Big_data)

The main distinguishing feature is the processing of a hitherto unprecedented amount of information and the development of methods for the effective processing of such data for a variety of purposes

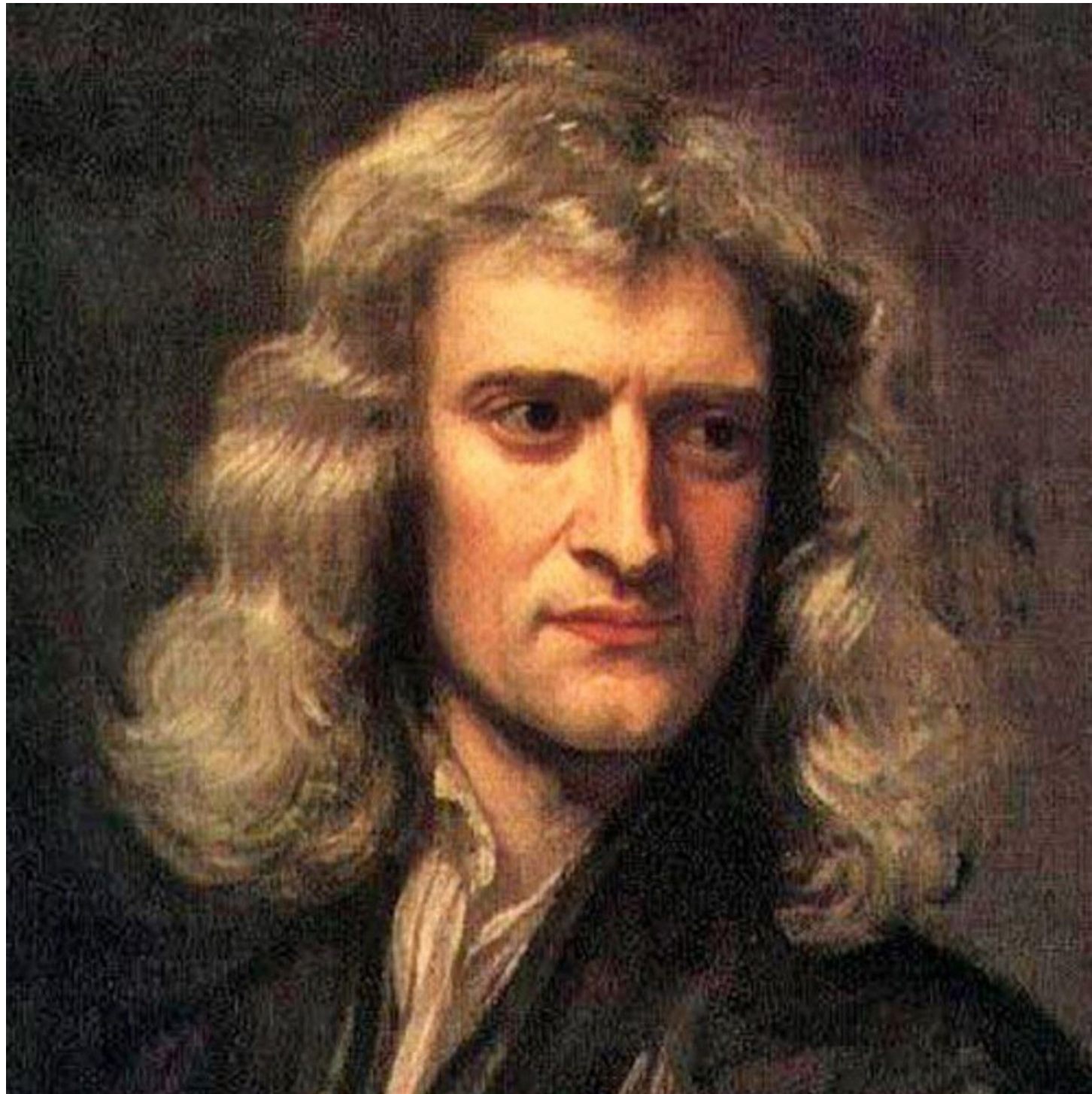
Relation of terms and spheres





Data → Knowledge

Long before the ML

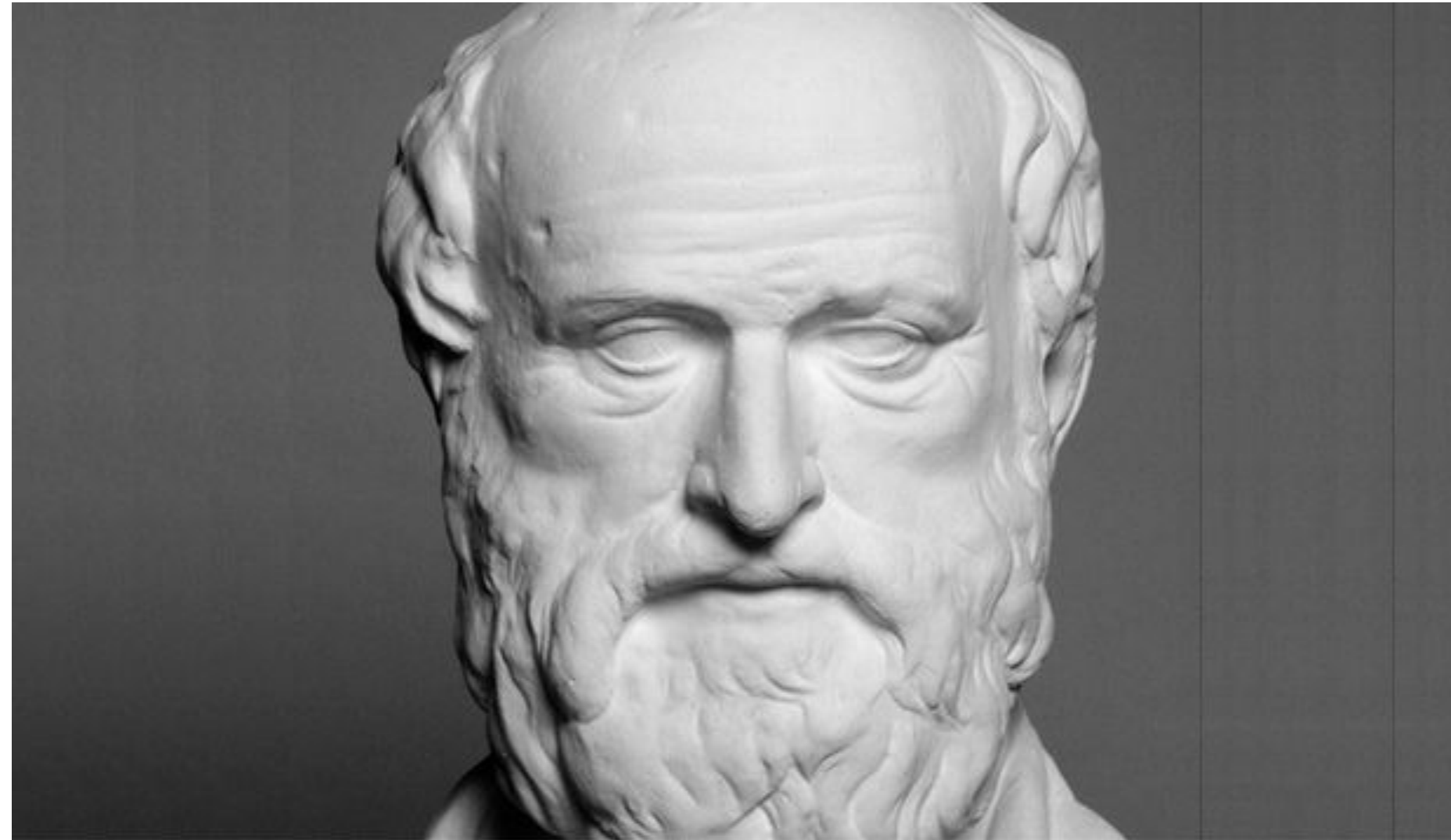


Isaac Newton



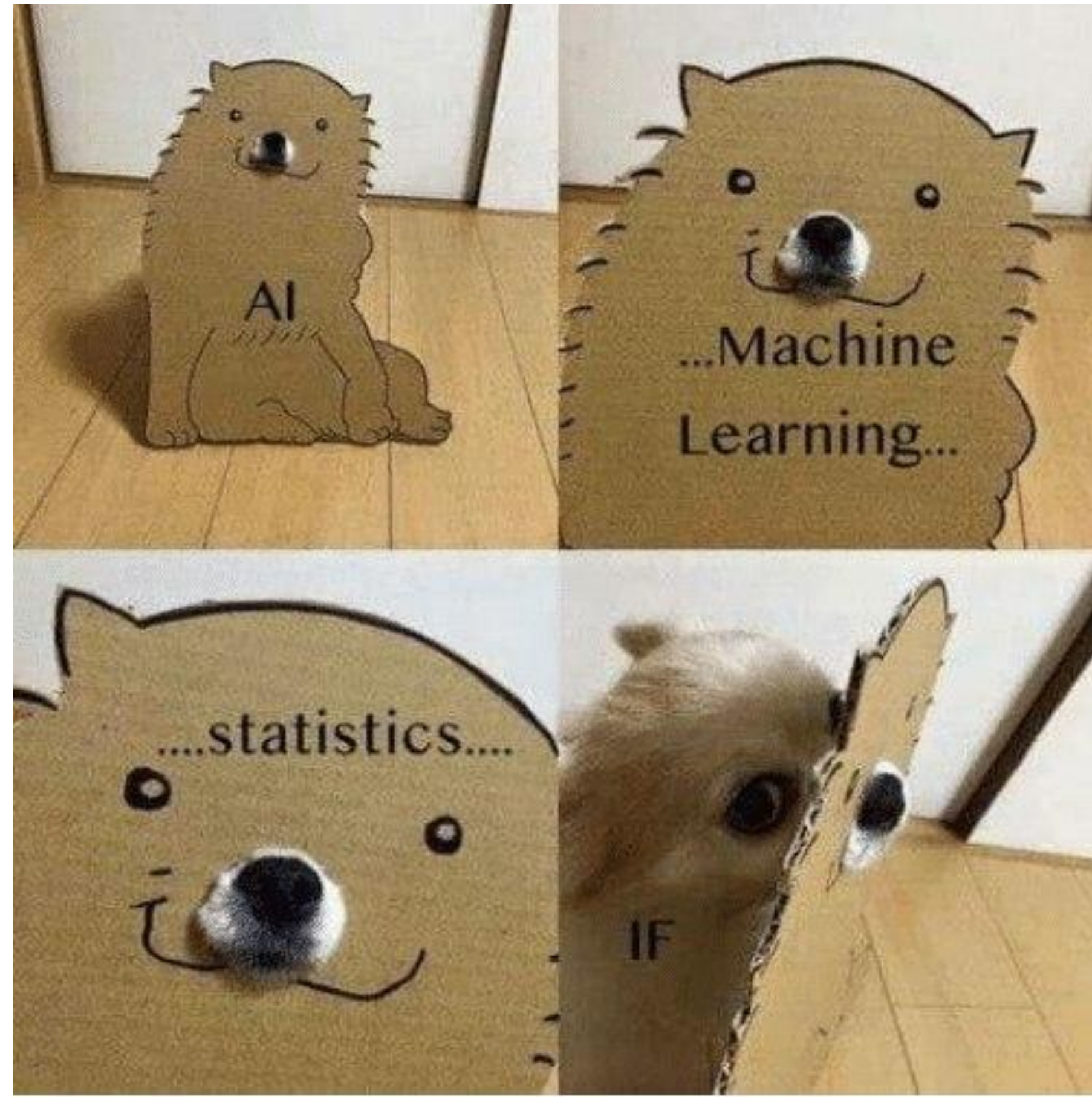
Johannes Kepler

Long before the ML



Eratosthenes

What it is actually



ML thesaurus

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ai

03

ML thesaurus



Denote the **dataset**.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



Observation (or datum, or data point) is one piece of information.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

In many cases the **observations** are supposed to be ***i.i.d.***

- ***independent***
- ***identically distributed***

ML thesaurus



Feature (or predictor, field) represents some special property.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



These all are features

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



These all are features

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



These all are features

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



These all are features

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



And even the name is a **feature**

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

ML thesaurus



The **feature matrix or design matrix** contains all the observations and their features

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

Features can even be multidimensional, we will discuss it later in this course

Matrix notation: features



Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

Feature matrix is usually denoted as $X \in R^{n \times p}$ (aka **design matrix**)

where n is number of objects in dataset and p is number of properties

ML thesaurus



Target represents the information we are interested in.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

Target can be either a **number** (real, integer, etc.) – for **regression** problem

ML thesaurus



Target represents the information we are interested in.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

Or a **label** – for **classification** problem

ML thesaurus



Target represents the information we are interested in.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Target (passed)
John	22	5	4	Brown	English	5	TRUE
Aahna	17	4	5	Brown	Hindi	4	TRUE
Emily	25	5	5	Blue	Chinese	5	TRUE
Michael	27	3	4	Green	French	5	TRUE
Some student	23	3	3	NA	Esperanto	2	FALSE

Mark can be treated as a label too (due to finite number of labels: 1 to 5)

Further here we will work with the numerical target (mark)



Matrix notation: target

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)
John	22	5	4	Brown	English	5
Aahna	17	4	5	Brown	Hindi	4
Emily	25	5	5	Blue	Chinese	5
Michael	27	3	4	Green	French	5
Some student	23	3	3	NA	Esperanto	2

Target matrix is usually denoted as $Y \in R^n$

where n is number of objects in dataset

ML thesaurus



The ***prediction*** contains values we predicted using some ***model***.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Predicted (mark)
John	22	5	4	Brown	English	5	4.5
Aahna	17	4	5	Brown	Hindi	4	4.5
Emily	25	5	5	Blue	Chinese	5	5
Michael	27	3	4	Green	French	5	3.5
Some student	23	3	3	NA	Esperanto	2	3

One could notice that prediction just averages of Statistics and Python marks. So our ***model*** can be represented as follows:

$$\text{mark}_{ML} = \frac{1}{2}\text{mark}_{Statistics} + \frac{1}{2}\text{mark}_{Python}$$

ML thesaurus



The **prediction** contains values we predicted using some **model**.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Predicted (mark)
John	22	5	4	Brown	English	5	1
Aahna	17	4	5	Brown	Hindi	4	5
Emily	25	5	5	Blue	Chinese	5	2
Michael	27	3	4	Green	French	5	4
Some student	23	3	3	NA	Esperanto	2	3

Different models can provide different predictions (even not meaningful):

$$\text{mark}_{ML}^{\hat{}} = \text{random}(\text{integer from } [1; 5])$$

ML thesaurus



The ***prediction*** contains values we predicted using some ***model***.

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Predicted (mark)
John	22	5	4	Brown	English	5	1
Aahna	17	4	5	Brown	Hindi	4	5
Emily	25	5	5	Blue	Chinese	5	2
Michael	27	3	4	Green	French	5	4
Some student	23	3	3	NA	Esperanto	2	3

Different models can provide different predictions.

*Usually some ***hypothesis*** lies beneath the model choice.*

ML thesaurus



Loss function measures the error rate of our model.

Square deviation	Target (mark)	Predicted (mark)
16	5	1
1	4	5
9	5	2
1	5	4
1	2	3

- **Mean Squared Error** (where $\mathbf{y}$ is vector of targets):

$$MSE(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2$$

ML thesaurus



Loss function measures the error rate of our model.

Absolute deviation	Target (mark)	Predicted (mark)
4	5	1
1	4	5
3	5	2
1	5	4
1	2	3

- **Mean Absolute Error** (where $\mathbf{y}$ is vector of targets):

$$MAE(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N} \|\mathbf{y} - \hat{\mathbf{y}}\|_1 = \frac{1}{N} \sum_i |y_i - \hat{y}_i|$$

ML thesaurus



To learn something, our **model** needs some degrees of freedom:

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Predicted (mark)
John	22	5	4	Brown	English	5	4.5
Aahna	17	4	5	Brown	Hindi	4	4.5
Emily	25	5	5	Blue	Chinese	5	5
Michael	27	3	4	Green	French	5	3.5
Some student	23	3	3	NA	Esperanto	2	3

$$\text{mark}_{ML}^{\hat{}} = w_1 \cdot \text{mark}_{Statistics} + w_2 \cdot \text{mark}_{Python}$$

ML thesaurus



To learn something, our **model** needs some degrees of freedom:

Name	Age	Statistics (mark)	Python (mark)	Eye color	Native language	Target (mark)	Predicted (mark)
John	22	5	4	Brown	English	5	4.447
Aahna	17	4	5	Brown	Hindi	4	4.734
Emily	25	5	5	Blue	Chinese	5	5.101
Michael	27	3	4	Green	French	5	3.714
Some student	23	3	3	NA	Esperanto	2	3.060

$$\text{mark}_{ML}^{\hat{}} = w_1 \cdot \text{mark}_{Statistics} + w_2 \cdot \text{mark}_{Python}$$

Main concepts



- Dataset (both X and Y)
 - Design matrix (only X)
 - Feature matrix (only X)
- Observation
 - object
 - datum
 - row of X
- Feature
 - column of X
- Target
 - Label
 - Y vector
- Model
- Prediction

Datasets and where to find them

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04

Datasets search



Nowadays there are tons of data available on the Internet.

It covers most of the cases you can think of.

So the main problem is to search for the right one!

Let's overview some ways to collect datasets.

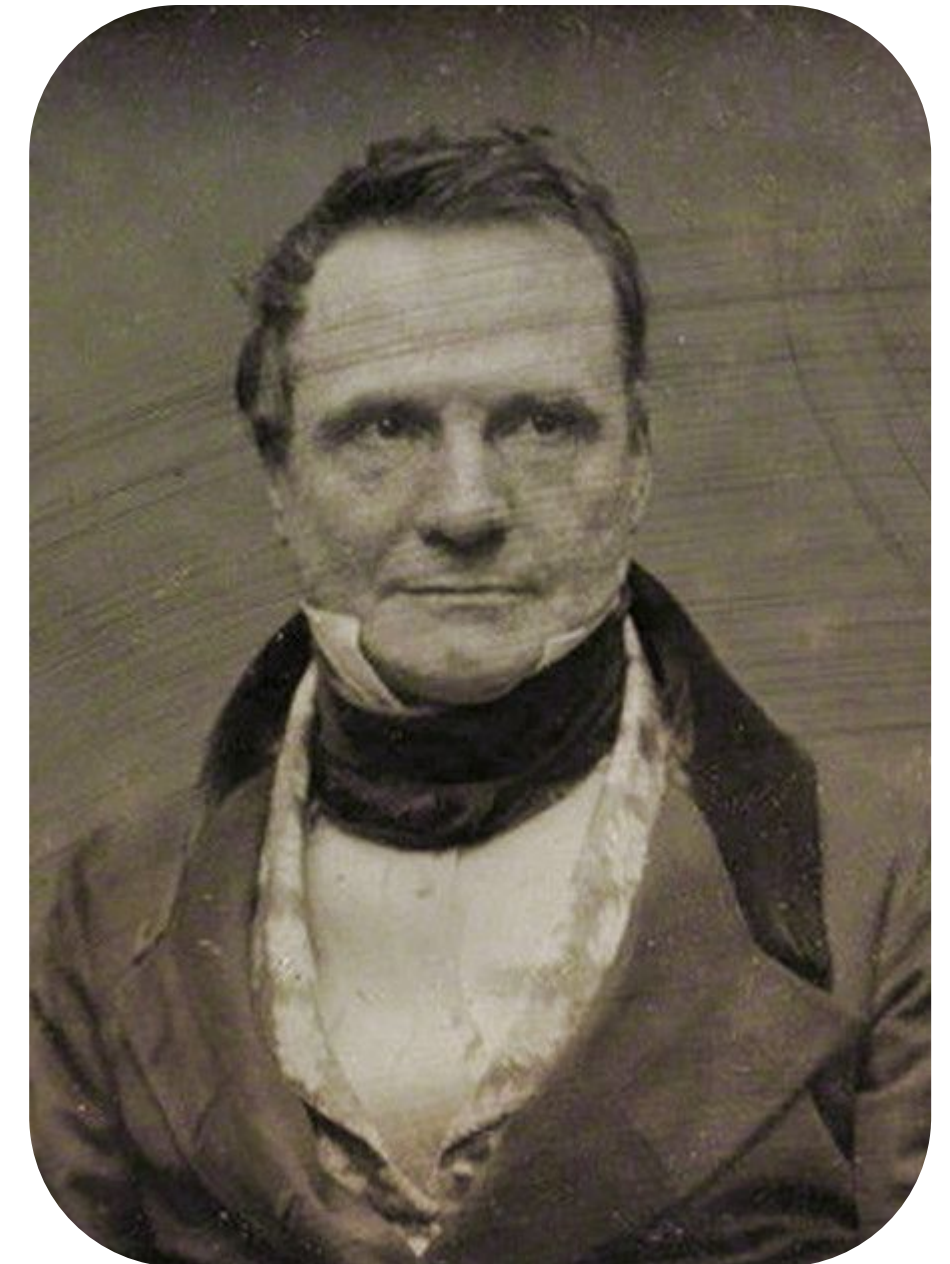
Garbage in, garbage out principle



is the concept that flawed, biased or poor quality ("garbage") information or input produces a result or output of similar ("garbage") quality.

On two occasions I have been asked, "Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?" ... I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question.

—Charles Babbage,
Passages from the Life of a Philosopher, 1864



https://en.wikipedia.org/wiki/Garbage_in%2C_garbage_out

Google dataset search



<https://datasetsearch.research.google.com/>

Contains main info about dataset
and links to sources.

Not all datasets are easily available

The screenshot shows the Google Dataset Search interface. At the top, the Google logo is on the left, and a search bar contains the text 'cancer cell segmentation'. Below the search bar are several filter buttons: 'Last updated', 'Download format', 'Croissant', 'Usage rights', 'Topic', 'Provider', and 'Free'. The results section shows '100+ data sets found'. The first result is highlighted with a grey background and features the Zenodo logo. The title of the first result is 'Cell Colony Image Segmentation Dataset 1 for T-47D Breast Cance...'. Below the title, it lists 'zenodo.org' and 'explore.openscience.eu' as sources, and 'txt, zip' as file formats. It also states 'Updated Mar 11, 2021'. To the right of this result, there is a detailed view of the same dataset. It shows the Zenodo logo, the full title 'Cell Colony Image Segmentation Dataset 1 for T-47D Breast Cancer Cells', and buttons to 'Explore at: zenodo.org' and 'explore.openscience.eu'. Below this, it lists 'txt, zip' as file formats. Further down, it provides a 'Unique identifier' (DOI: https://doi.org/10.5281/zenodo.4593510), the date the 'Data set updated' (Mar 11, 2021), the 'Dataset provided by' (Zenodo), the 'Authors' (Delmon Arous; Stefan Schrunner; Ingunn Hanson; Nina F.J. Edin; Eirik Malinen), and the 'Licence' (Attribution 4.0 (CC BY 4.0)).

Google

cancer cell segmentation

▼ Last updated ▼ Download format Croissant ▼ Usage rights ▼ Topic ▼ Provider Free

100+ data sets found

Cell Colony Image Segmentation Dataset 1 for T-47D Breast Cance...
zenodo.org
explore.openscience.eu
txt, zip
Updated Mar 11, 2021

Cell Colony Image Segmentation Dataset 1 for T-47D Breast Cancer Cells
Explore at: [zenodo.org](#) [explore.openscience.eu](#)
txt, zip

Unique identifier
<https://doi.org/10.5281/zenodo.4593510>

Data set updated
Mar 11, 2021

Dataset provided by
[Zenodo](#)

Authors
Delmon Arous; Stefan Schrunner; Ingunn Hanson; Nina F.J. Edin; Eirik Malinen; Delmon Arous; Stefan Schrunner; Ingunn Hanson; Nina F.J. Edin; Eirik Malinen

Licence
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Licence information was derived automatically

Kaggle

<https://www.kaggle.com/datasets>



Datasets

Explore, analyze, and share quality data. [Learn more](#) about data types, creating, and collaborating.

+ New Dataset

Your Work

Search datasets

Filters

All datasets

Computer Science

Education

Classification

Computer Vision

NLP

Data Visualization

Pre-Trained Model

Trending Datasets

See All

51 - 100	Moderate
101 - 150	Unhealthy for S
151 - 200	Unhealthy

2023 Air Quality Data for CBSAs

Nikki Perry · Updated 20 hours ago
Usability **10.0** · 166 kB

16

Smite Item Statistics Data

Matt OP · Updated a day ago
Usability **10.0** · 15 kB
1 File (CSV)

16

Machine Learning Engineer Salary in 2024

Chopper53 · Updated 21 hours ago
Usability **10.0** · 110 kB
1 File (CSV)

8

FuelConsumption

Krupa Dharamshi · Updated a month ago
Usability **10.0** · 6 kB
1 File (CSV)

15

Hugging Face



<https://huggingface.co/datasets>

 **Hugging Face**

Models Datasets Spaces Posts Docs Solutions Pricing Log In [Sign Up](#)

Main Tasks Libraries Languages Licenses Other

Modalities

3D Audio Geospatial Image

Tabular Text Time-series Video

Size (rows)

< 1K > 1T

Format

json csv parquet imagefolder

soundfolder webdataset text arrow

Datasets 223,270

google/frames-benchmark

Viewer • Updated 1 day ago • 824 • 840 • 126

FBK-MT/mosel

Viewer • Updated 6 days ago • 51.1M • 120 • 47

glaiveai/reflection-v1

Viewer • Updated 19 days ago • 60.1k • 428 • 39

k-mktr/improved-flux-prompts-photoreal-portrait

Viewer • Updated 5 days ago • 20k • 131 • 65

nvidia/OpenMathInstruct-2

Viewer • Updated about 17 hours ago • 22M • 173 • 29

argilla/FinePersonas-v0.1

Viewer • Updated 20 days ago • 21.1M • 399 • 306

migtissera/Synthia-v1.5-I

fka/awesome-chatgpt-prompts

Viewer • Updated Sep 4 • 170 • 8.42k • 5.83k

HackerNoon/where-startups-trend

Preview • Updated 8 days ago • 70 • 38

KingNish/reasoning-base-20k

Viewer • Updated 3 days ago • 19.9k • 178 • 35

openai/MMMLU

Viewer • Updated 5 days ago • 393k • 6.21k • 377

lmms-lab/LLaVA-Video-178K

Viewer • Updated 2 days ago • 1.63M • 146 • 34

5CD-AI/Viet-LAION-Gemini-VQA

Viewer • Updated 5 days ago • 844k • 149 • 25

40umov/dostoevsky

62

Papers with code (RIP)



<https://paperswithcode.com/datasets>

Datasets

9,761 machine learning datasets

Share your dataset with the ML community!

Best match

Filter by Modality

Images	2661
Texts	2546
Videos	862
Audio	394
Medical	336
3D	303

9761 dataset results

CIFAR-10 (Canadian Institute for Advanced Research, 10 classes)
The CIFAR-10 dataset (Canadian Institute for Advanced Research, 10 classes) is a subset of the Tiny Images dataset and consists of 60000 32x32 color images. The images are labelled...
14,073 PAPERS • 98 BENCHMARKS

ImageNet
The ImageNet dataset contains 14,197,122 annotated images according to the WordNet hierarchy. Since 2010 the dataset is used in the ImageNet Large Scale Visual Recognition...
13,418 PAPERS • 40 BENCHMARKS

MS COCO (Microsoft Common Objects in Context)
The MS COCO (Microsoft Common Objects in Context) dataset is a large-scale object detection, segmentation, key-point detection, and captioning dataset. The dataset consists of...
10,128 PAPERS • 92 BENCHMARKS

UCI ML repository



<https://archive.ics.uci.edu/>

Historical source of datasets.

Most of them collected in 90s and 00s.

We use some of them in this course.

Welcome to the UC Irvine Machine Learning Repository

We currently maintain 664 datasets as a service to the machine learning community. Here, you can donate and find datasets used by millions of people all around the world!

[VIEW DATASETS](#) [CONTRIBUTE A DATASET](#)

Popular Datasets	New Datasets
Iris A small classic dataset from Fisher, 1936. One of the earliest known datasets used for... Classification 150 Instances 4 Features	PhiUSIIL Phishing URL (Website) PhiUSIIL Phishing URL Dataset is a substantial dataset comprising 134,850 legitimate ... Classification 235.8K Instances 54 Features
Heart Disease 4 databases: Cleveland, Hungary, Switzerland, and the VA Long Beach Classification 303 Instances 13 Features	RT-IoT2022 The RT-IoT2022, a proprietary dataset derived from a real-time IoT infrastructure, is in... Classification, Regres... 123.12K Instances 84 Features
Dry Bean Images of 13,611 grains of 7 different registered dry beans were taken with a high-res... Classification 13.61K Instances 16 Features	Regensburg Pediatric Appendicitis This repository holds the data from a cohort of pediatric patients with suspected app... Classification 782 Instances 59 Features
Rice (Cammeo and Osmancik) A total of 3810 rice grain's images were taken for the two species, processed and feat... Classification 3.81K Instances 7 Features	National Poll on Healthy Aging (NPHA) This is a subset of the NPHA dataset filtered down to develop and validate machine le... Classification 714 Instances 15 Features

Chatbots



Nowadays it is very good idea to give [some chatbot](#) a task to perform deep research on datasets availability on your topic for you.

Example for [3D cancer segmentation](#).

- Perplexity
 - best for web search
- ChatGPT
 - the big guy
- ...

You are top machine learning researcher who stays in the field for at least 30 years, publishing papers and participating in discussions in top conferences such as NIPS, ICML, ICCV, SIGIR, WSDM, etc. You are especially interested in biological imaging field.

Give an overview of the most impactful contemporary datasets for 3D cancer detection and segmentation. Include foundational datasets of past as well as the biggest and most valuable datasets of today.

Thought for 2m 44s >

Short version: the 3D cancer-imaging universe today is dominated by a small pantheon of CT/MRI/PET datasets (LIDC/LUNA, BraTS, LiTS, KiTS, HECKTOR, AutoPET, MSD, PROMISE, NSCLC-*), with a newer wave of whole-body and 3D microenvironment resources starting to appear. Let's walk through them by organ/site and also by historical "waves".

0. Meta-infrastructure

Most of the impactful 3D cancer datasets live in three ecosystems:

- **TCIA (The Cancer Imaging Archive)** – key host for LIDC-IDRI, NSCLC Radiomics/Radiogenomics, NCI-ISBI prostate, AutoPET, QIN collections, etc. ScienceDirect

Standardization initiative

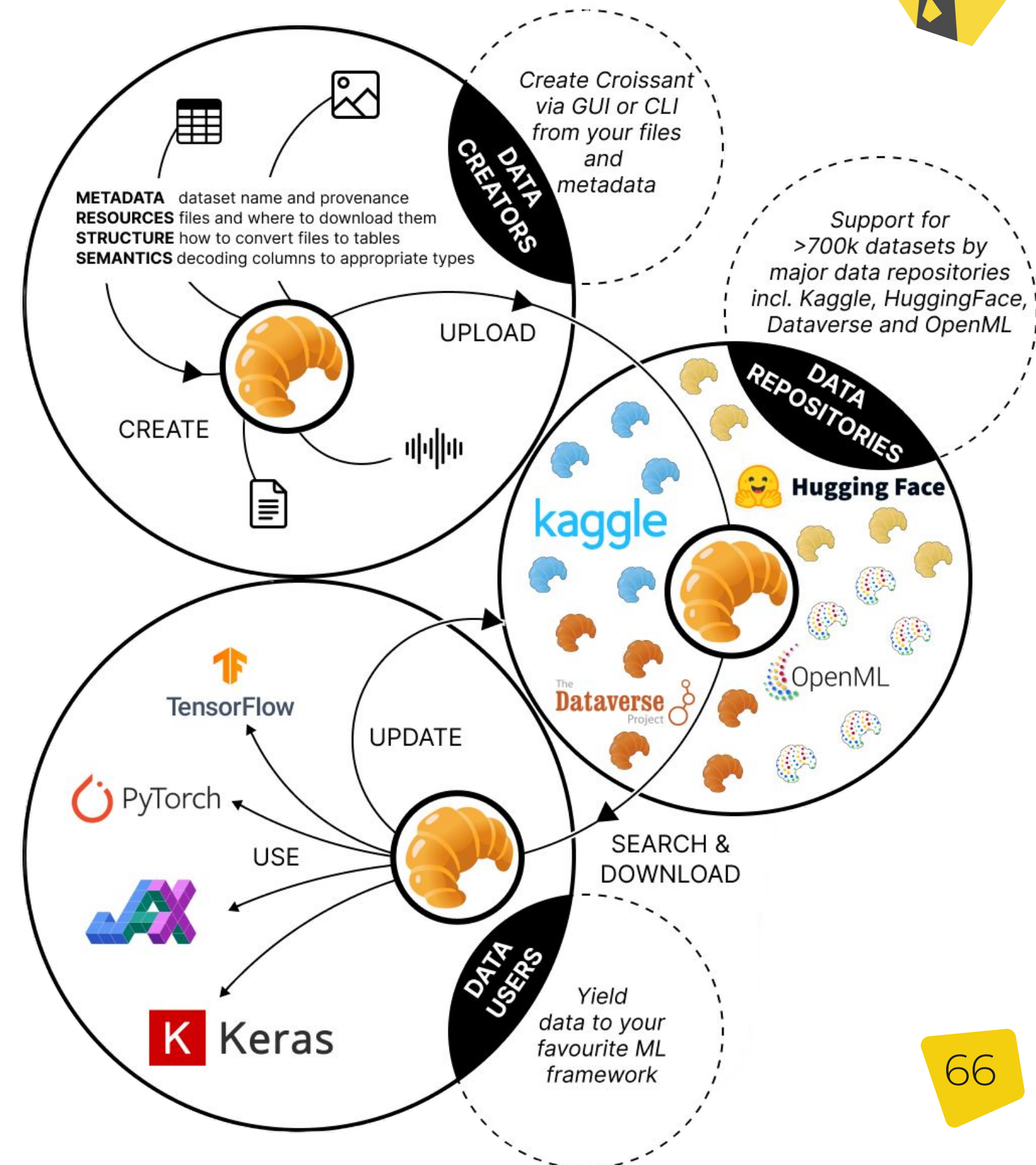


Croissant dataset standard

<https://mlcommons.org/working-groups/data/croissant/>

<https://research.google/blog/croissant-a-metadata-format-for-ml-ready-datasets/>

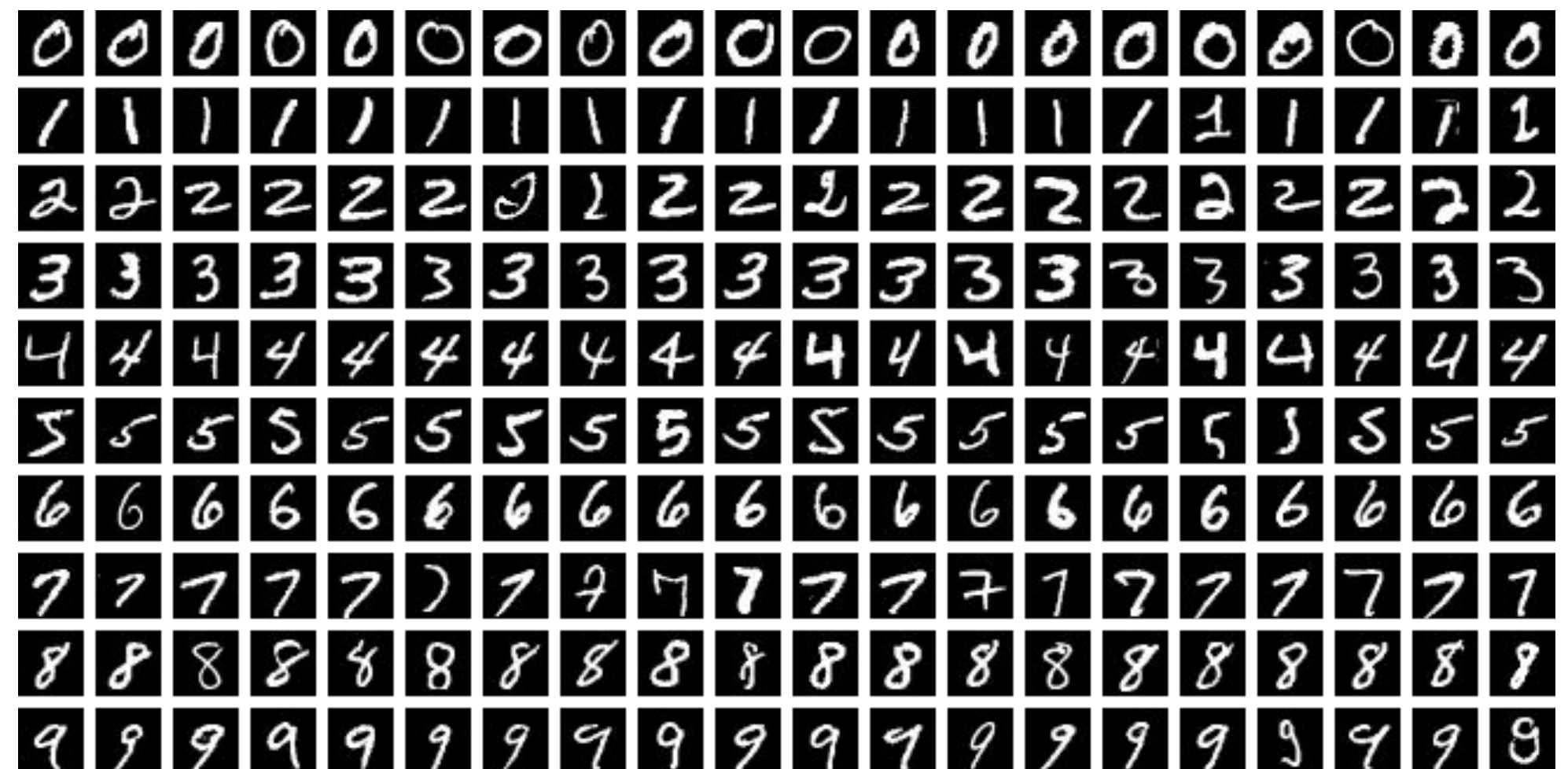
<https://github.com/mlcommons/croissant>



Remarkable datasets



- Fisher's irises
- Boston house prices (cancelled)
- Breast cancer
- MNIST
 - 1990s
 - 10k of 28*28 images grayscale
 - 10 classes
 - notMNIST
 - FashionMNIST
- ImageNet
 - 2010s
 - 14m of 224*224 images
 - 1000 classes



ML competitions



<https://mlcontests.com/state-of-machine-learning-competitions-2024/>

Exploratory data analysis (EDA)

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05

Dataset analysis

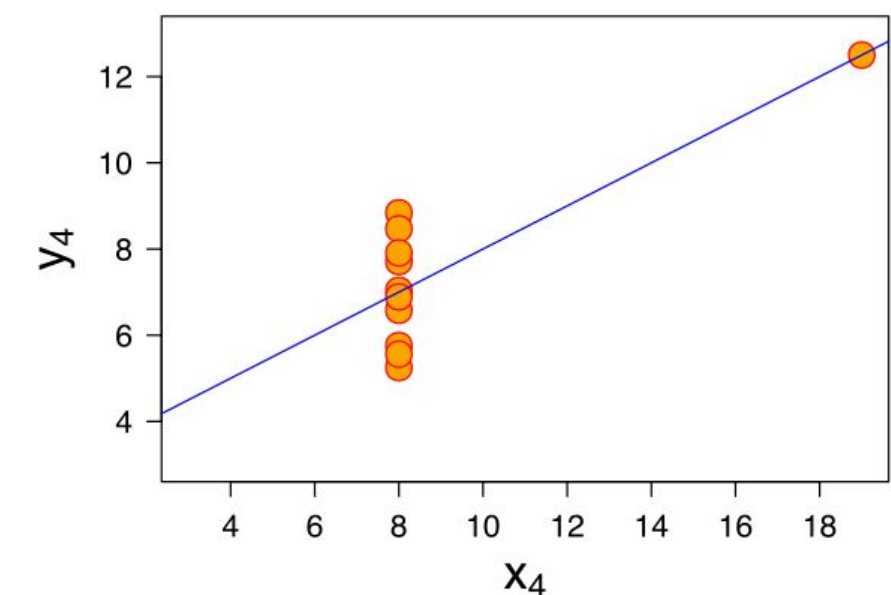
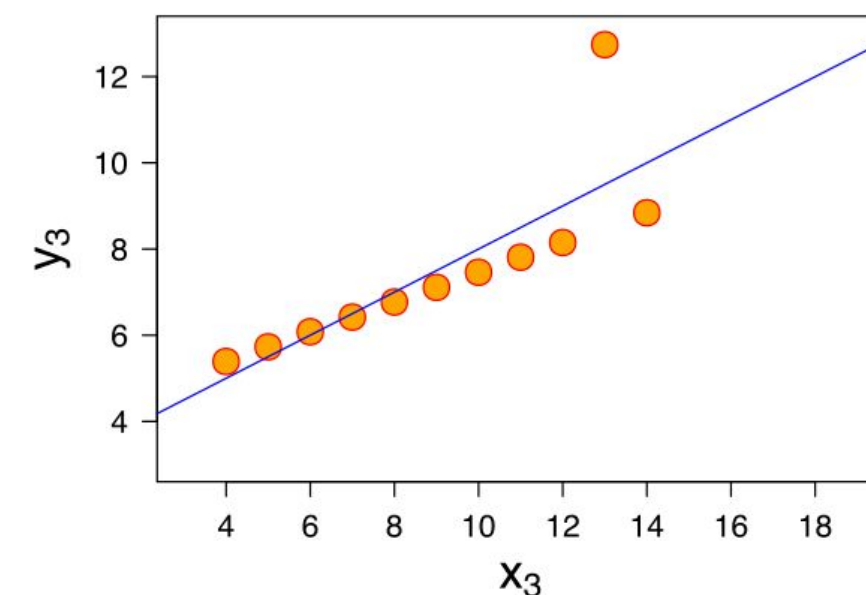
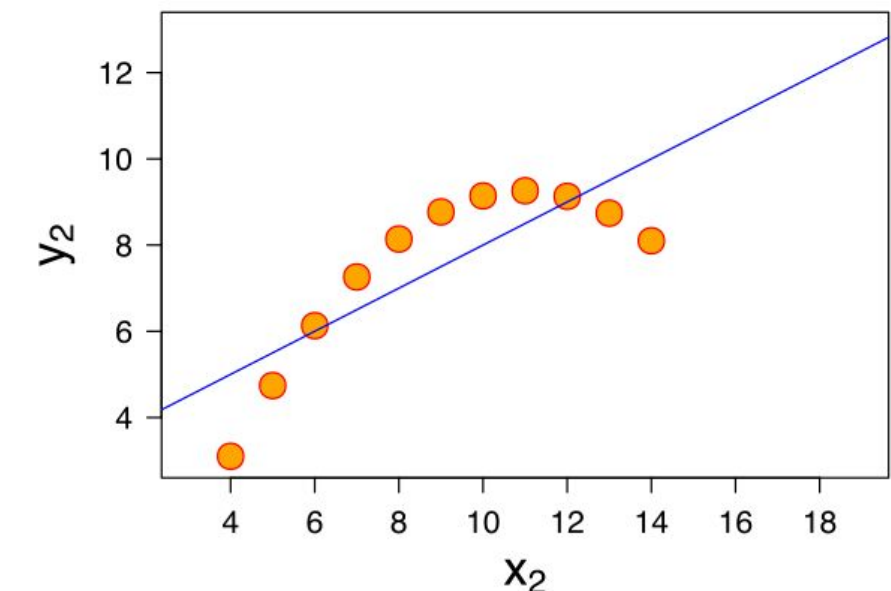
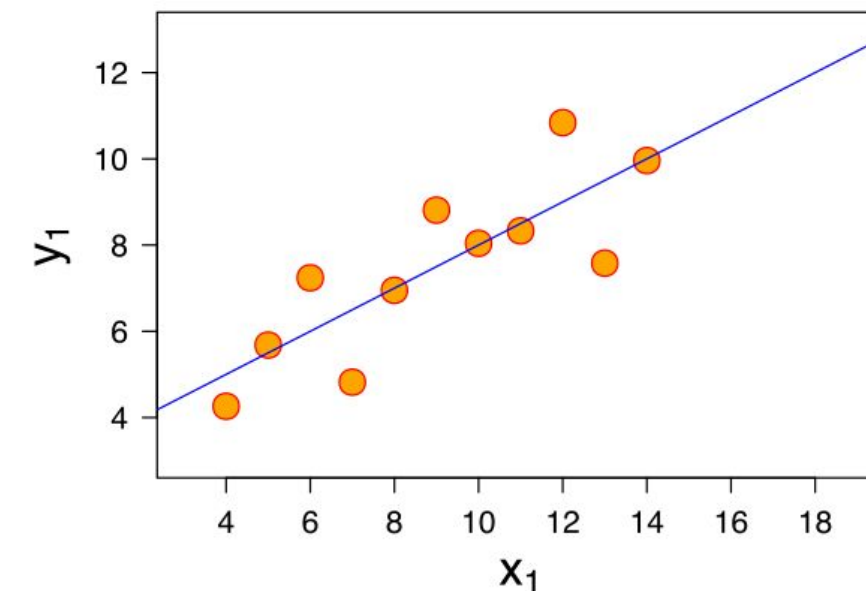


Before you build any model it's important to analyse data you have with various methods because all the models depend on data and its structure.

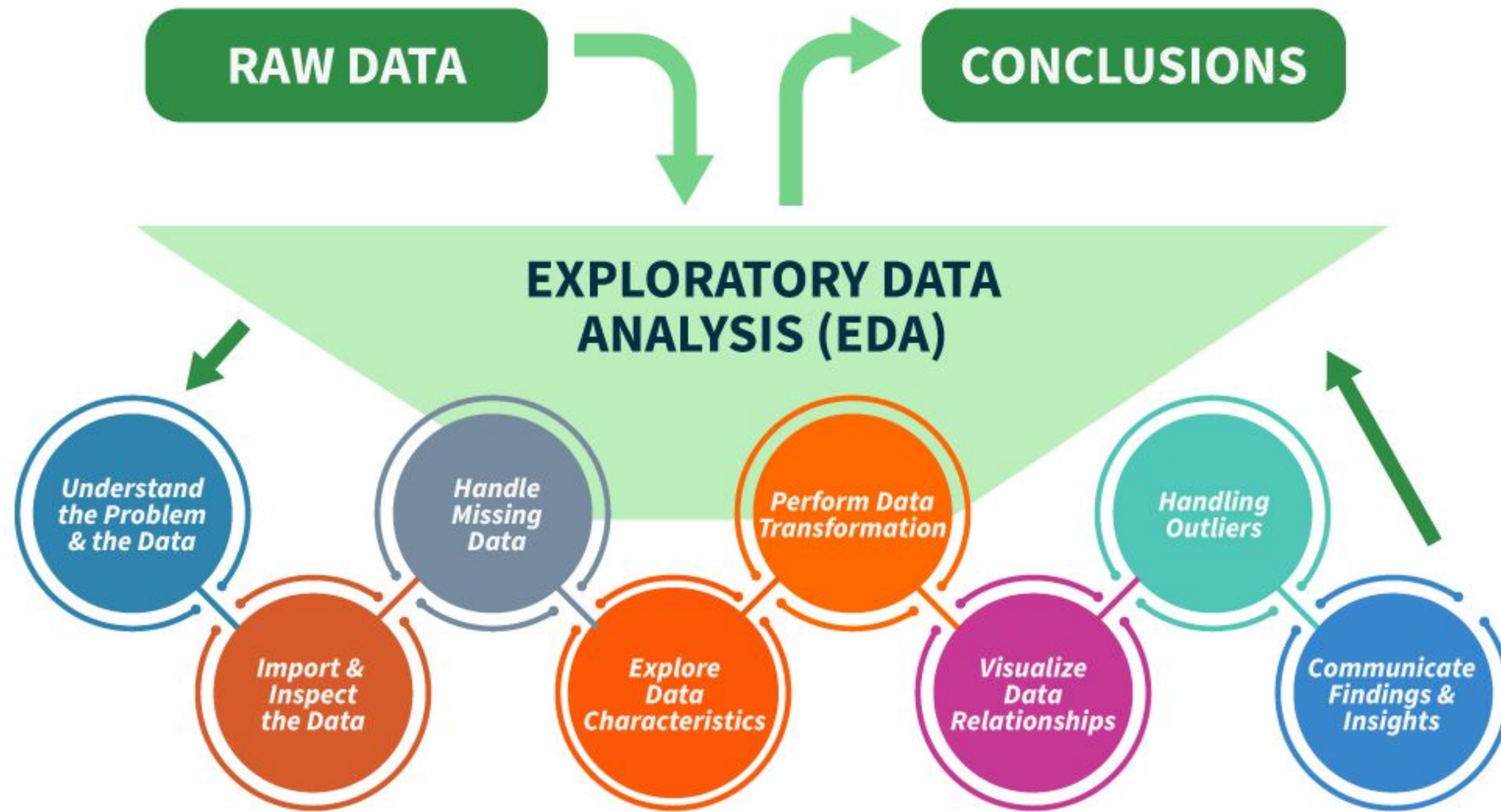
See [Anscombe's quartet](#) for example:

four datasets that have nearly identical simple descriptive statistics, yet have very different distributions and appear very different when graphed

This process is called
Exploratory Data Analysis (EDA)



Steps of EDA



Categorical features encoding



Great overview

https://github.com/Dyakonov/PZAD/blob/master/2020/PZAD2020_042featureengineering_07.pdf

Whole ebook on feature engineering <https://feaz-book.com/>

One hot encoding



Maximum Likelihood Estimation

girafe
ai

06

Parametric and nonparametric models



Nonparametric statistics is a type of statistical analysis that makes minimal assumptions about the underlying distribution of the data being studied. Often these models are infinite-dimensional, rather than finite dimensional, as is parametric statistics.

Nonparametric statistics can be used for descriptive statistics or statistical inference. Nonparametric tests are often used when the assumptions of parametric tests are evidently violated.

[© Common knowledge site](#)

Likelihood maximization



Consider the most simple case of discrete features and target.

Denote dataset X, Y generated by distribution with parameter θ

Likelihood of a parameter is defined as probability of sampling this particular data in case underlying distribution is defined by this parameter.

Maximization of likelihood means we choose the most probable parameters having this particular dataset

$$L(\theta|X, Y) = P(X, Y|\theta) \rightarrow \max_{\theta}$$

Note that likelihood is not probability function of θ

i.i.d. property



We can employ i.i.d property of data samples to split probability of the whole dataset into independent problems

$$P(X, Y|\theta) = \prod_i P(x_i, y_i|\theta)$$

Then we apply logarithm function to both parts of equation above

$$\log P(X, Y|\theta) = \sum_i \log P(x_i, y_i|\theta)$$

The latter expression is easier to operate with:
later we will predict log-probability of each object directly

[Importance of iid and example of non iid data](#)

Log-likelihood equivalence



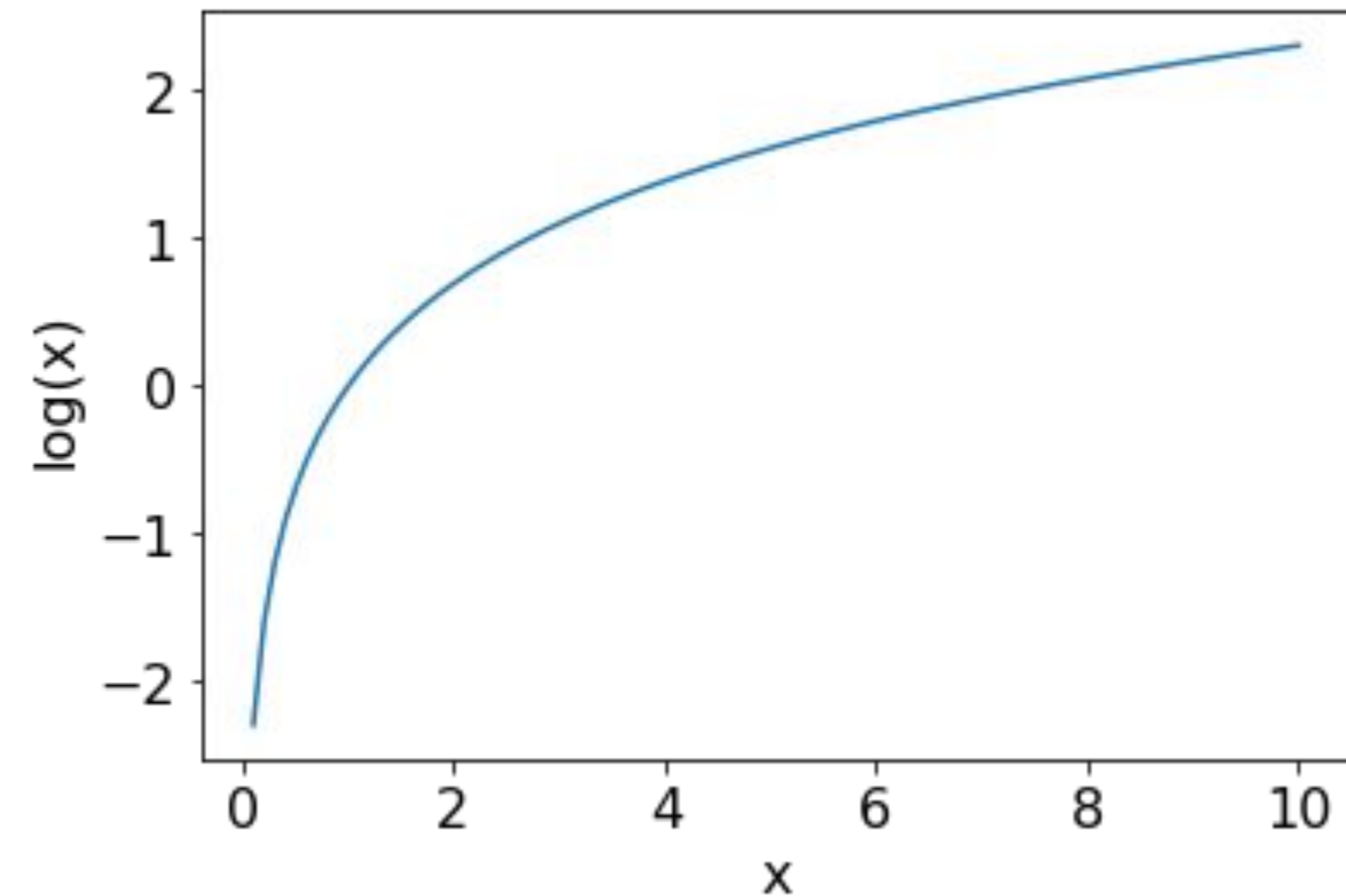
Since logarithm is a convex function on an open set, it preserves maximum of expression when applied, so that

$$L(\theta|X, Y) \rightarrow \max_{\theta}$$

and

$$\log L(\theta|X, Y) \rightarrow \max_{\theta}$$

have the same solutions in terms of θ



Maximum Likelihood formula



$$\hat{\theta} = \arg \max_{\theta} L(\theta|X, Y)$$

is called maximum likelihood estimation of model parameters.

In optimization theory functions are usually minimized, so the same problem could be reformulated using **Negative Log-Likelihood (NLL)** loss

$$\hat{\theta} = \arg \min_{\theta} - \sum_i \log P(x_i, y_i | \theta)$$

Continuous parameter set



Here we formulate MLE in terms of probability which means we assume finite number of parameter values.

Defining MLE for infinite parameters set is left as easy exercise for you.
(Mostly replace summation with integral)

$$\hat{\theta} = \arg \min_{\theta} - \sum_i \log P(x_i, y_i | \theta)$$



Not only MLE

Generally in statistics plenty of ways to estimate parameters exist:

- Maximum likelihood estimators
- Bayes estimators
- Method of moments estimators
- Cramér–Rao bound
- Maximum a posteriori (MAP or EM algorithm)
- Particle filter
- Markov chain Monte Carlo (MCMC)
- Kalman filter, and its various derivatives
- Wiener filter

see https://en.wikipedia.org/wiki/Estimation_theory#Estimators

<https://en.wikipedia.org/wiki/Estimator>

https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm

Useful materials on MLE



- Nice video explanation
 - <https://www.youtube.com/watch?v=YevSE6bRhTo>
- Textual intro to MLE
 - <https://www.statlect.com/fundamentals-of-statistics/maximum-likelihood>
 - <https://www.statlect.com/fundamentals-of-statistics/maximum-likelihood-algorithm>

Machine Learning problems overview

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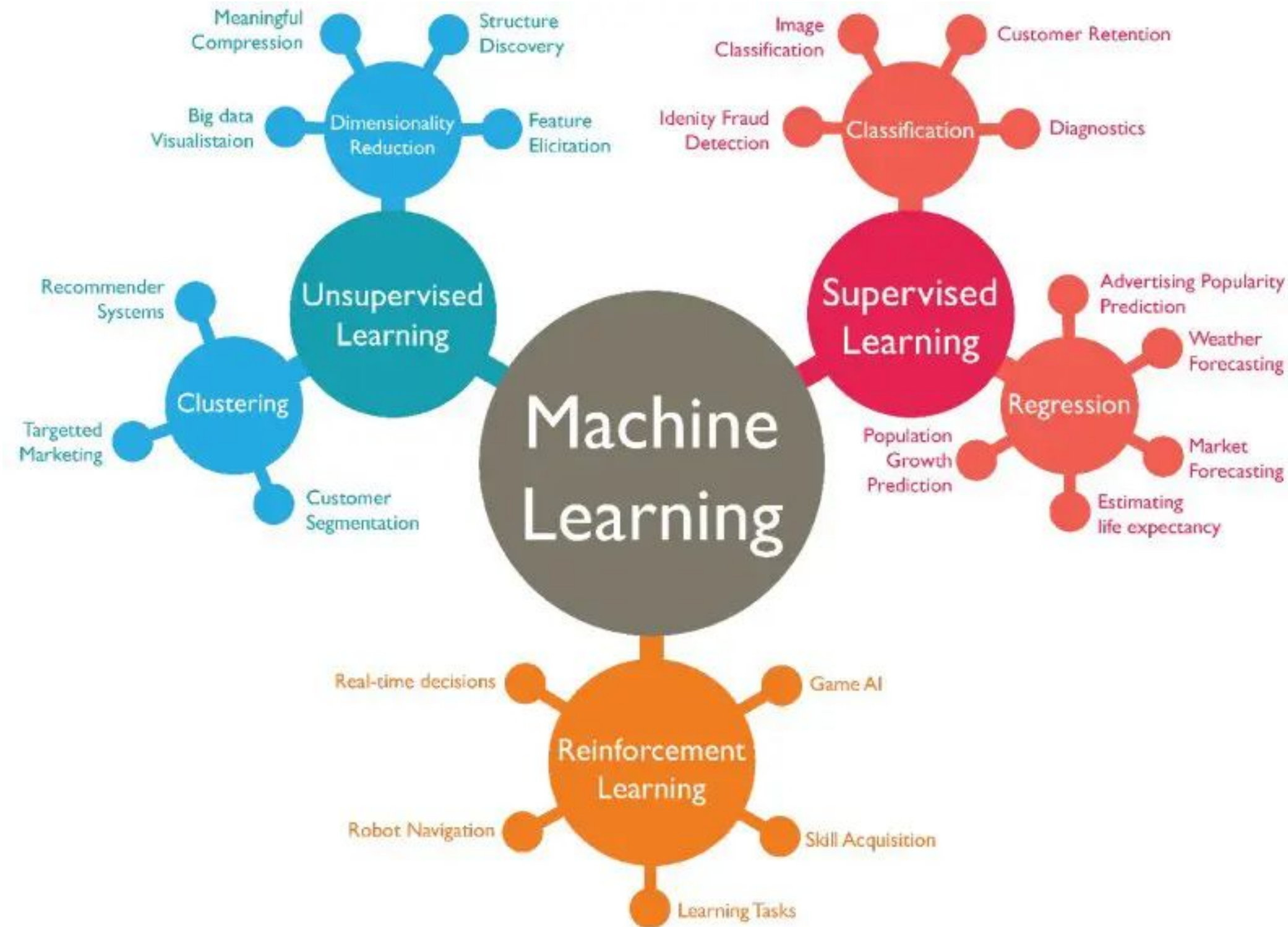
07

Taxonomy of ML problems



- Supervised
 - true answers are given
- Unsupervised
 - no true answers
- Semi-supervised
 - transductive inference
- Self-supervised
- Reinforced
- Generative
- etc...

There's no full and non-intersecting taxonomy of ML tasks, but terms above are used in practice



Main entities



Problem statement

In $\rightarrow$ Out

Satisfies $\uparrow$

Dataset
(aka sample)

Training data

$X Y$

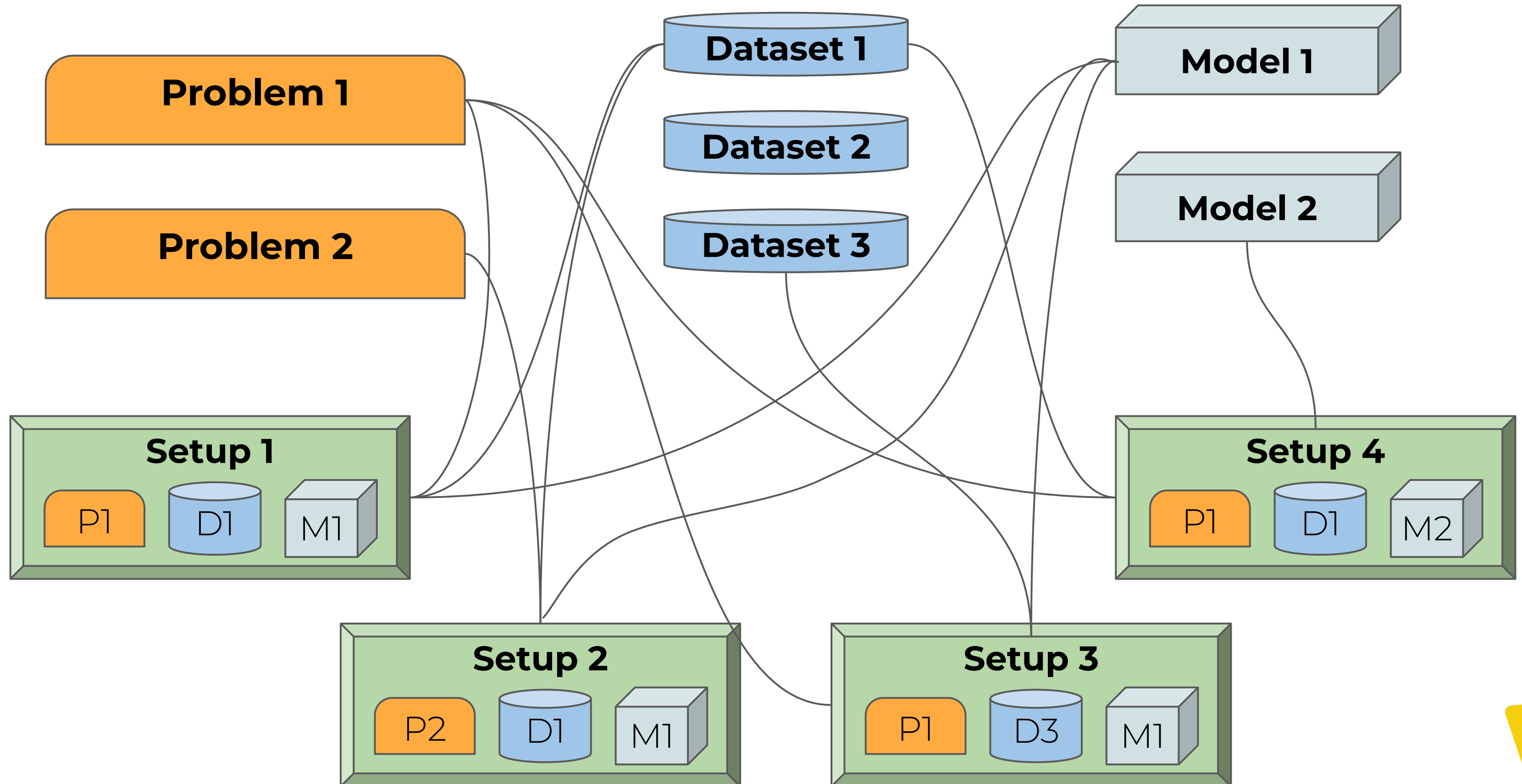
$\uparrow$ Implements

Model

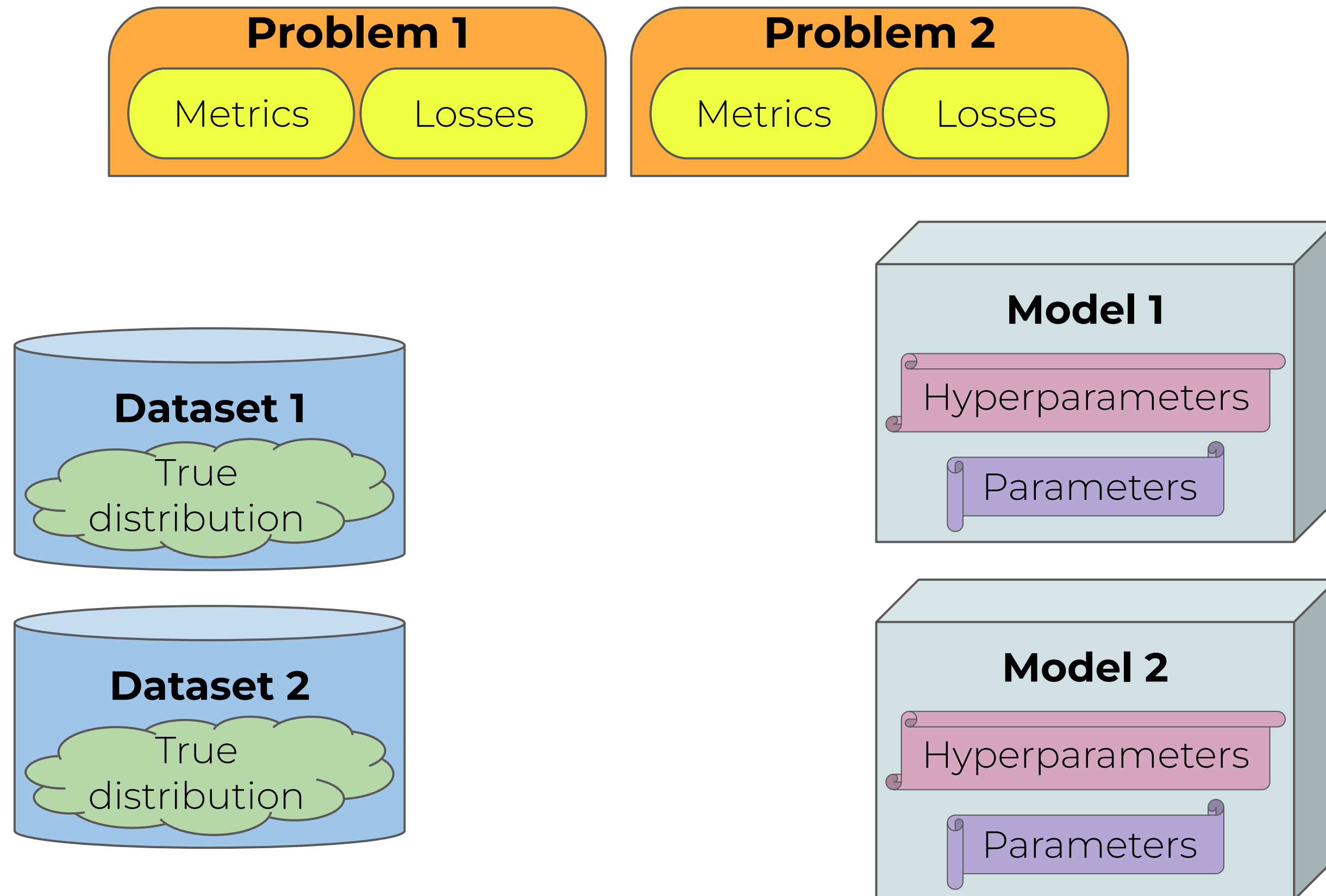
True distribution

Test data
 X

Practical problem setup



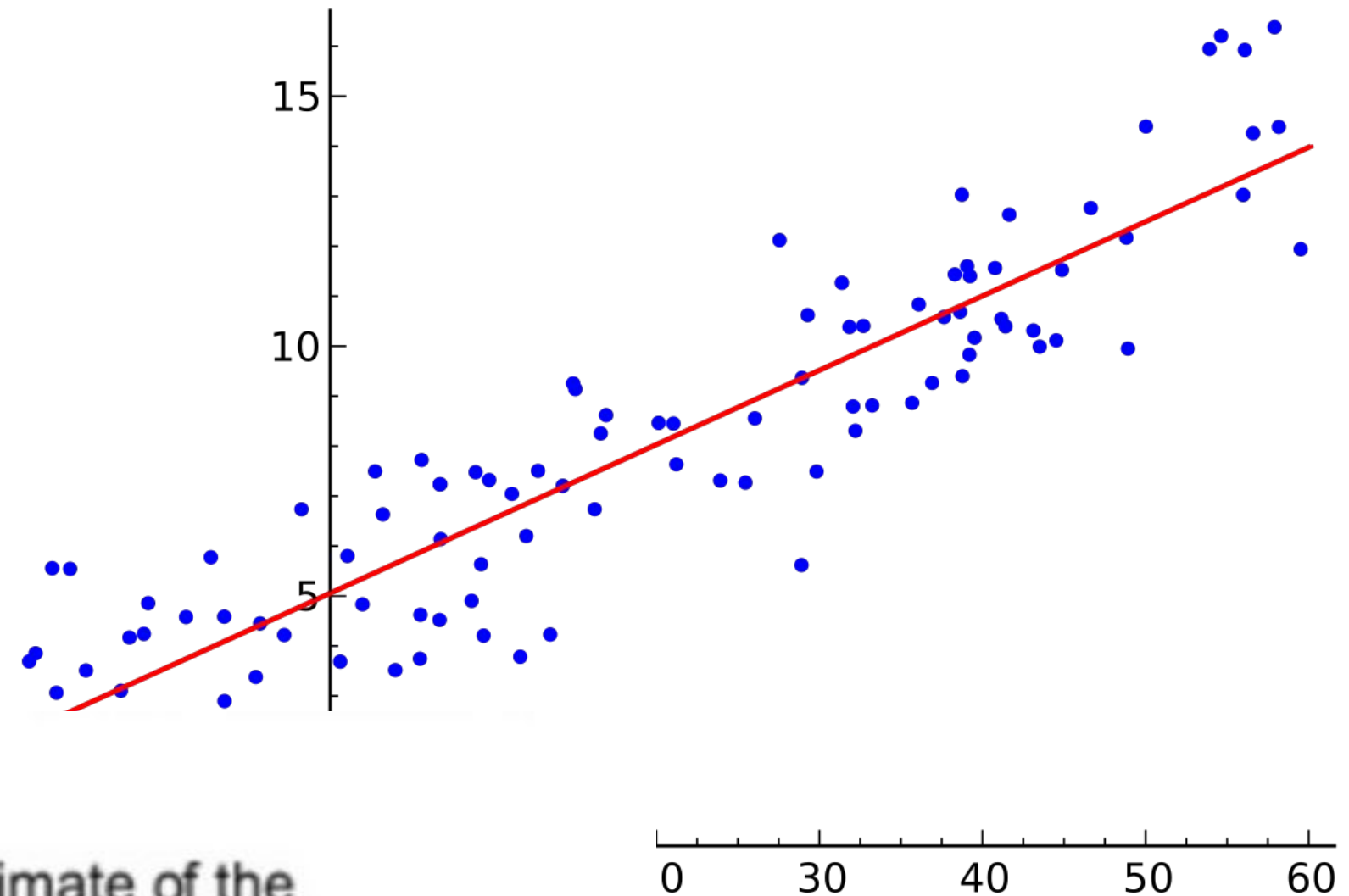
Practical problem components



Problems covered in this course



- Regression problem



Estimated
(or predicted)
Y value for
observation i

Estimate of
the regression
intercept

Estimate of the
regression slope

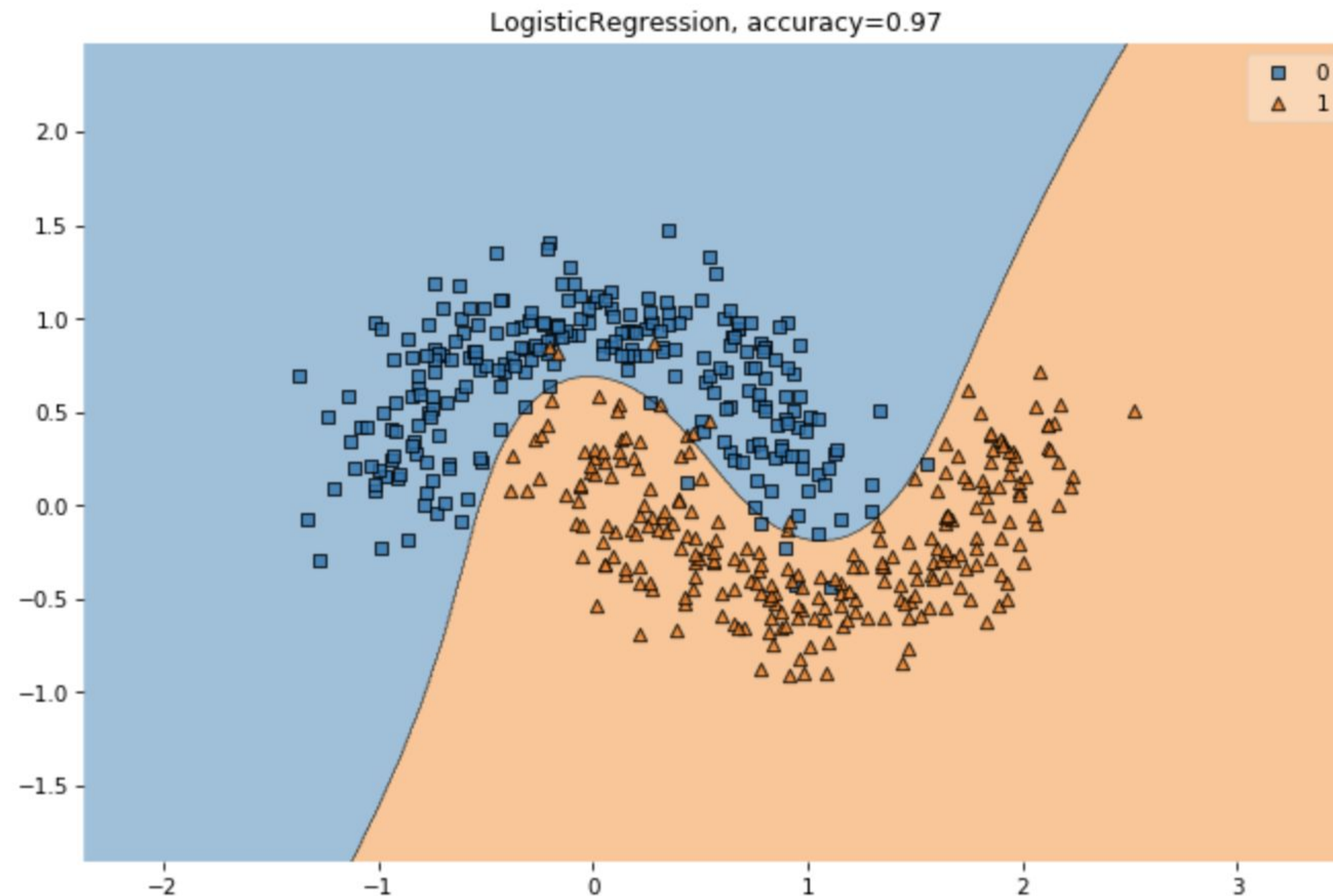
Value of X for
observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

Problems covered in this course



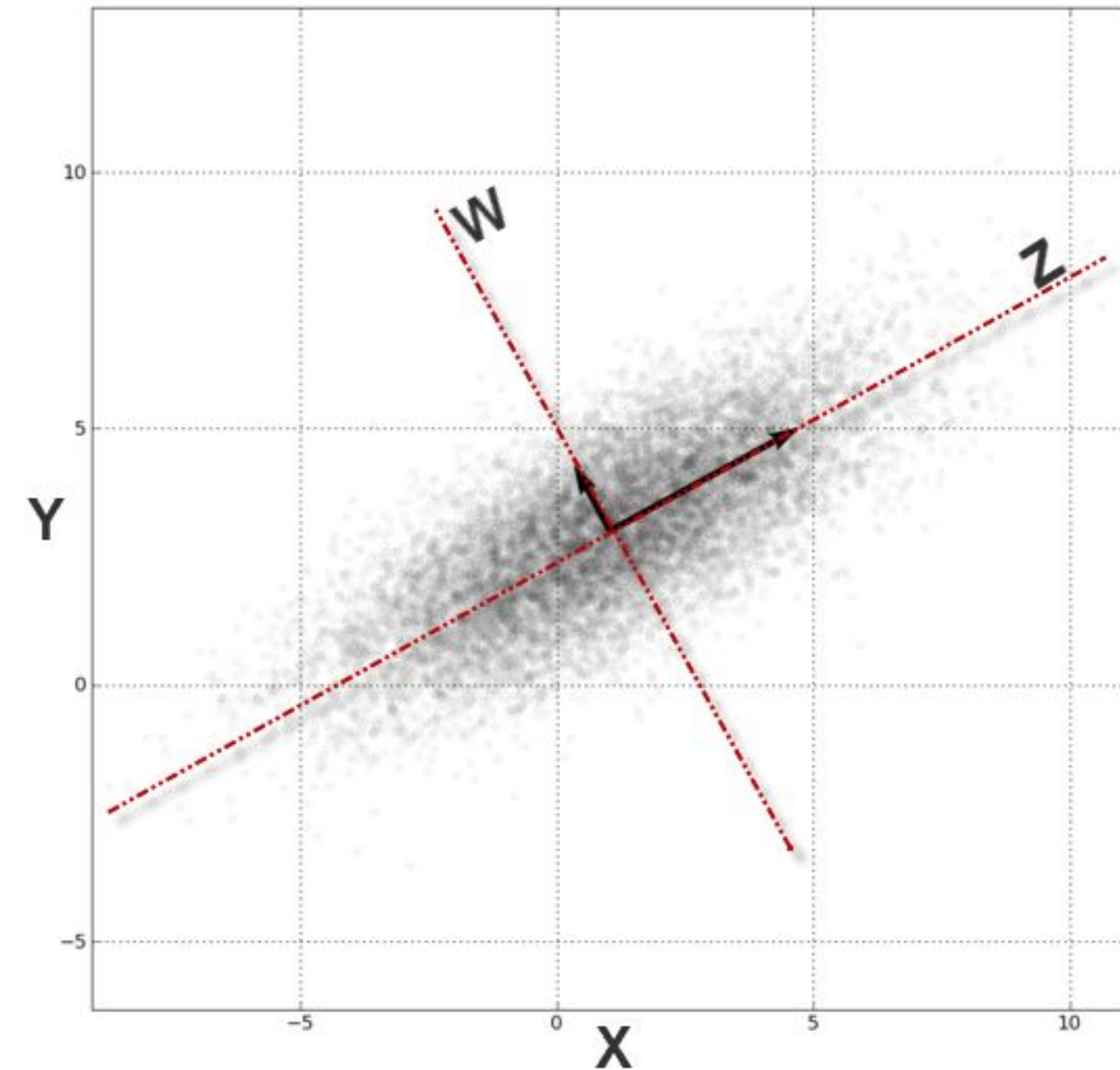
- Regression problem
- Classification problem



Problems covered in this course



- Regression
- Classification
- Dimensionality reduction
- Clustering
- Anomaly detection
- Generation



Naïve Bayes classifier

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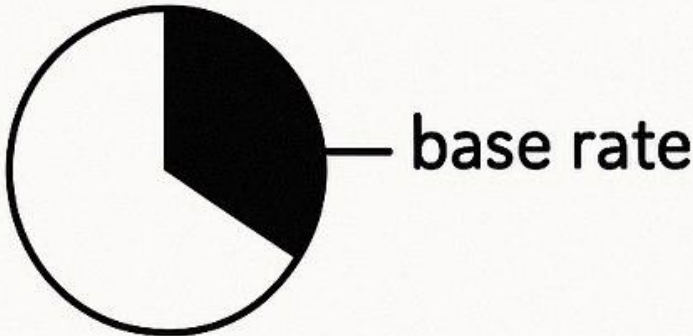
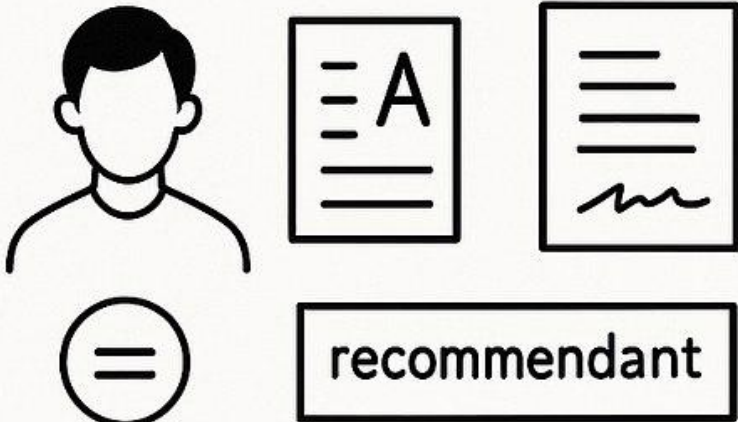
08



Bayesian statistics

is built around idea of distributions (not outcomes) as a central entity.

Short bayesian statistics intro: [part 1](#), [part 2](#)

FREQUENTIST	BAYESIAN
• Probability = long-run frequency • Parameters are fixed • Confidence intervals What is the probability that a student chooses mathematics at university in the United States? 	• Probability = degree of belief • Parameters have a distribution • Posterior distribution What is the probability that a student chooses mathematics in the United States? 

Bayes' theorem



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

or, in our case

$$P(y_i = C_k | \mathbf{x}_i) = \frac{P(\mathbf{x}_i | y_i = C_k)P(y_i = C_k)}{P(\mathbf{x}_i)}$$

Naïve Bayes classifier



$\mathbf{x}_i \in \mathbb{R}^p$ $y_i \in \{C_1, \dots, C_K\}$ for K-class classification

$$P(y_i = C_k | \mathbf{x}_i) = \frac{P(\mathbf{x}_i | y_i = C_k) P(y_i = C_k)}{P(\mathbf{x}_i)}$$

Naïve assumption: **features are *independent***

Naive assumption



$$P(y_i = C_k | \mathbf{x}_i) = \frac{P(\mathbf{x}_i | y_i = C_k) P(y_i = C_k)}{P(\mathbf{x}_i)}$$

features are independent

$$P(\mathbf{x}_i | y_i = C_k) = \prod_{l=1}^p P(x_i^l | y_i = C_k)$$

Naïve Bayes prediction



$$P(y_i = C_k | \mathbf{x}_i) = \frac{P(\mathbf{x}_i | y_i = C_k) P(y_i = C_k)}{P(\mathbf{x}_i)}$$

Optimal class label:

$$C^* = \arg \max_k P(y_i = C_k | \mathbf{x}_i)$$

To find maximum we even do not need the denominator

But we need it to get probabilities

Continuous features



to deal with them we need to introduce parametric likelihood of the features we decompose. Then we fit parameters on training set and use them to predict density for particular point.

Typical distributions include:

- Gaussian
- Multinomial
- Bernoulli

$$P(x_i \mid y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

Practical example



No	Statistics	Python	Target: Machine Learning
1	5	5	5
2	5	3	5
3	3	4	4
4	4	3	4
5	4	5	3
6	2	3	3

Class	Prob
5	0.33
4	0.33
3	0.33

feature	feature value	target	P
Statistics	2	5	0
Statistics	3	5	0
Statistics	4	5	0
Statistics	5	5	1

Machine learning libraries

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09

Sklearn



is the biggest library of ML methods implemented in one place.

Despite it is great tool for learning, it is not made for production
e.g. see <https://alexanderdyakonov.wordpress.com/2021/03/04/ml-scikit-learn/>

During this course we will give you tools which can be used for contemporary problems.

Sklearn is not made for production!!! Don't use for real life tasks!!!!

Pandas



is very handfull tool but it is quite slow.

For better experience you can get alternatives:

- Polars - **highly recommended!**
 - <https://stackoverflow.com/questions/77881942/whats-the-polars-equivalent-to-the-pandas-iloc-method>
- dask
- cudf
- swifter



How to store data?

Straightforward solution is a .csv file. But it has many disadvantages such as filesize, sequential reading and others.

Better alternatives:

- [parquet](#) - **recommended**
 - supported by pandas and others
 - column based
- [feather](#) - column based
- [ORC](#)
- [hdf5](#) - scientific format for datasets

References

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ai**

10

Great resources on machine learning



- [ru] <https://alexanderdyakonov.wordpress.com/>
- [ru] <https://github.com/esokolov/ml-course-hse>
- [en] <https://cs231n.stanford.edu/>
- [ru] <https://education.yandex.ru/handbook/ml>
- [en, ru] <https://github.com/rasbt/python-machine-learning-book-3rd-edition>
- [en, ru] <https://mlcourse.ai/book/index.html>
- [en] <https://datatalks.club/events.html>
- <https://www.classcentral.com/>
 - huge collection of open courses
- [en] <https://www.youtube.com/@statquest/videos>
 - <https://statquest.org/video-index/>
- [en] <https://ml-visualized.com/>

Books



- Pattern Recognition and Machine Learning; Christopher Bishop
- Machine Learning: A probabilistic perspective; Kevin Murphy
- Machine Learning; Tom Mitchell
- The Elements of Statistical Learning: Data Mining, Inference and Prediction; Trevor Hastie, Robert Tibshirani, Jerome Friedman
- [High-Dimensional Probability](#); Roman Vershynin
- **Introduction to Machine Learning**, Laurent Younes
<https://arxiv.org/abs/2409.02668>
- [Evaluating Learning Algorithms. A Classification Perspective](#); Nathalie Japkowicz, Mohak Shah
- Alice's Adventures in a differentiable wonderland <https://arxiv.org/pdf/2404.17625>
- Information Geometry and Its Applications, Shun-ichi Amari
<https://content.e-bookshelf.de/media/reading/L-7505366-733c6681c3.pdf>
- Математические основы теории машинного обучения и прогнозирования; В. В. Вьюгин <http://iitp.ru/upload/publications/6256/vyugin1.pdf>
- [Машинное обучение: основы](#); Николенко С. И

Takeouts



- To find parameters use MLE
- Remember the i.i.d. property
- The first dimension of a dataset corresponds to the number of objects
- The second (and so on) to the features/time/...
- Even the naïve assumptions may be suitable in some cases
- Simple models provide great baselines

Revise

1. ML and AI overview
2. Thesaurus and notation
3. Maximum Likelihood Estimation
4. Some Machine Learning problems
 - a. Classification
 - b. Regression
 - c. Dimensionality reduction
5. Naïve Bayes classifier

Thanks for attention!

Questions?



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